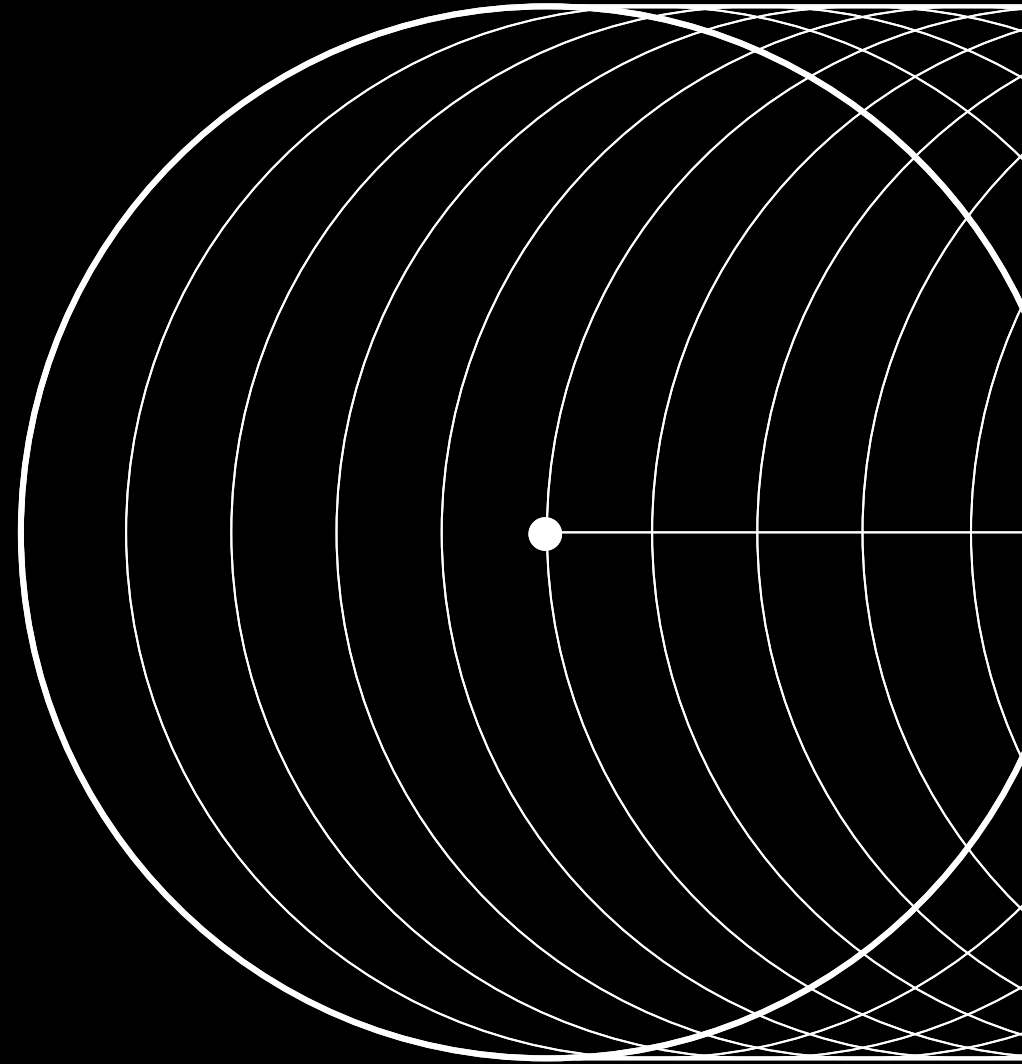


Yandex Research

Evolution of Conditioning and Multi-modal Generation



Sergey Kastrayulin



Agenda

- 01 Basic conditioning
- 02 Text and Image conditioning
- 03 Multi-modal generation
- 04 Inference-time compute scaling

Basic Conditioning

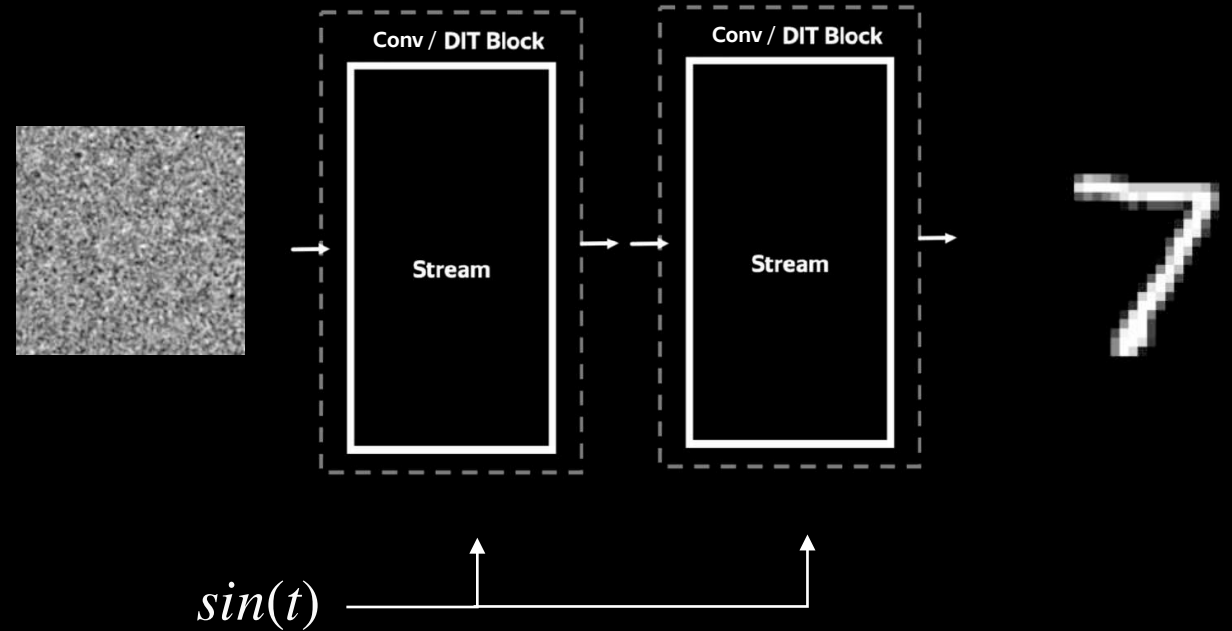


Unconditional

Basic Conditioning



Unconditional



Basic Conditioning

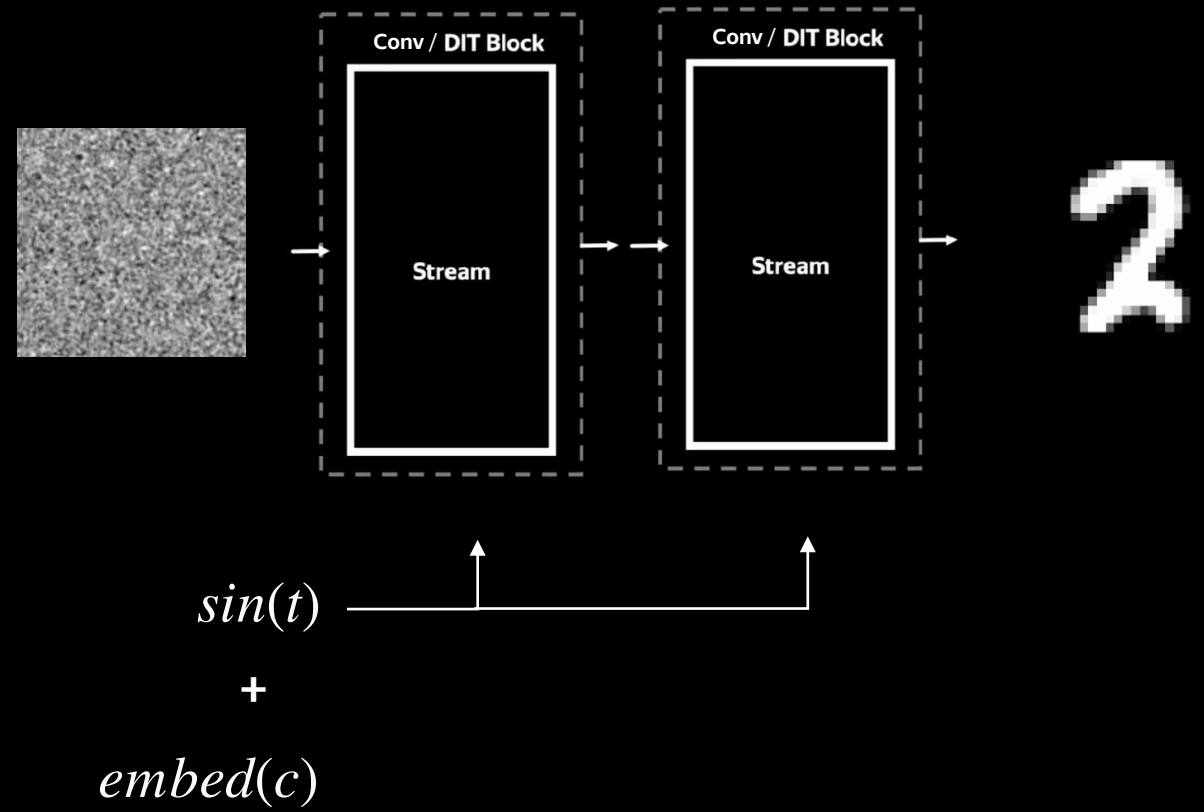
0	1	2	3	4	5
0	1	2	3	4	5
0	1	2	3	4	5
0	1	2	3	4	5
0	1	2	3	4	5
0	1	2	3	4	5
0	1	2	3	4	5

Class-conditional

Basic Conditioning



Class-conditional



Basic Conditioning

“Cyberpunk girl”



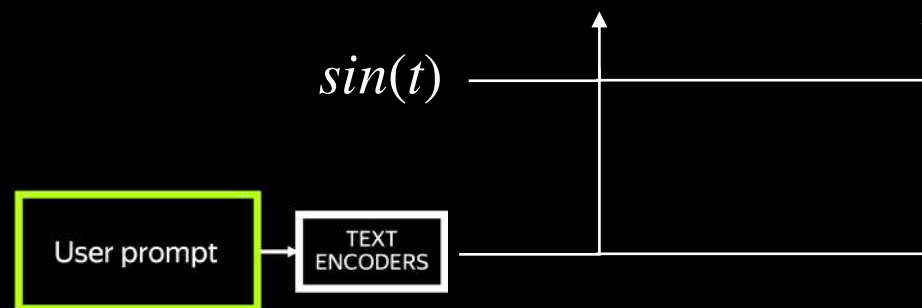
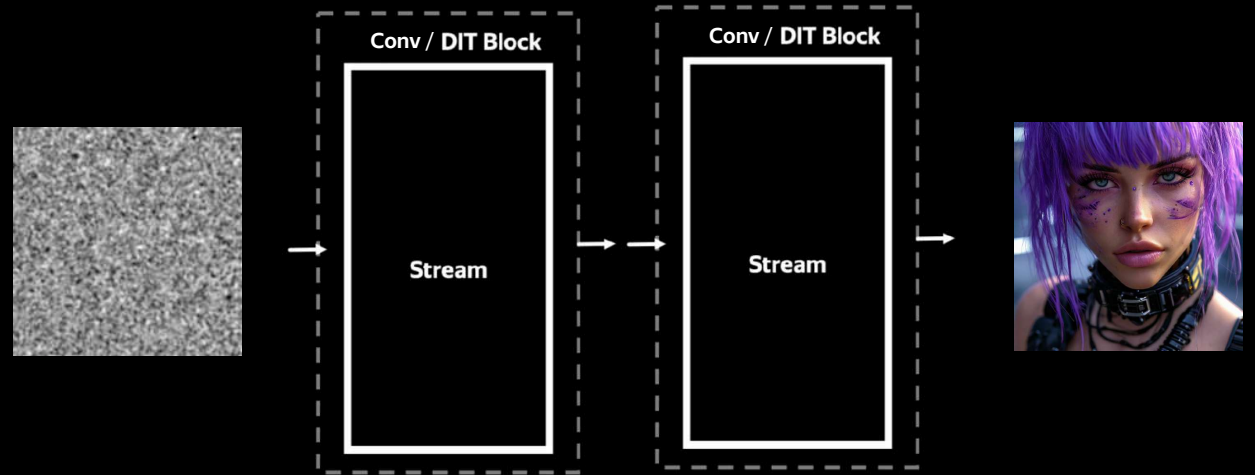
Text-conditional

Basic Conditioning

“Cyberpunk girl”

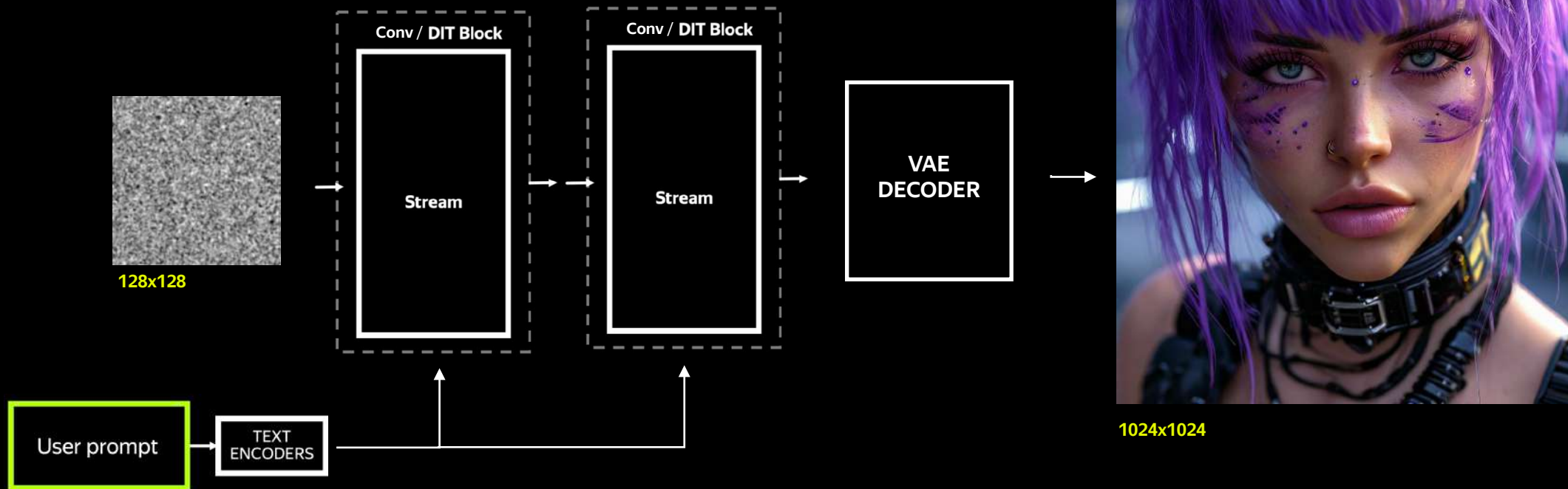


Text-conditional



Practical Consideration

Latent Diffusion



Conditioning on text and image



Restore photo



Add costume



Conditioning on text and image



Restore photo



Add costume



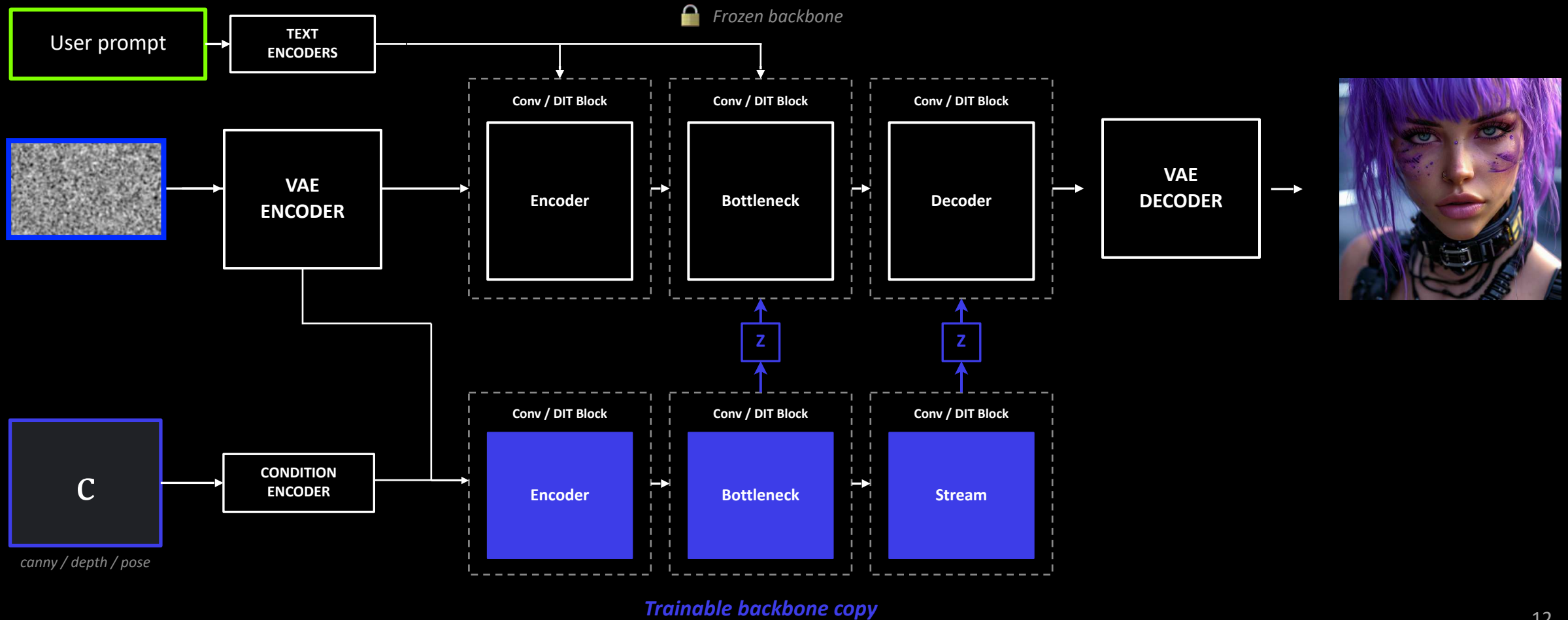
Why is it harder?

Preservation:

- Spatial/structural control
- Semantic/identity control

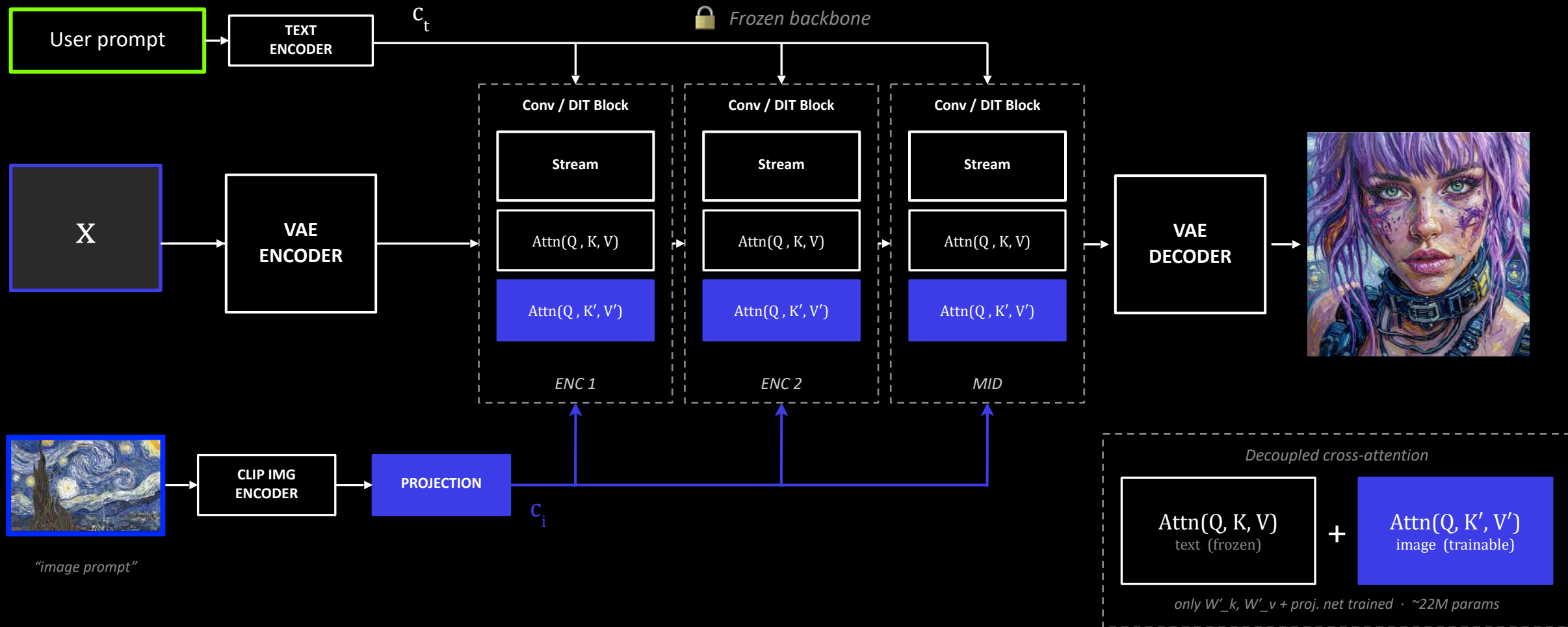
ControlNet

Spatial/Structural Conditioning

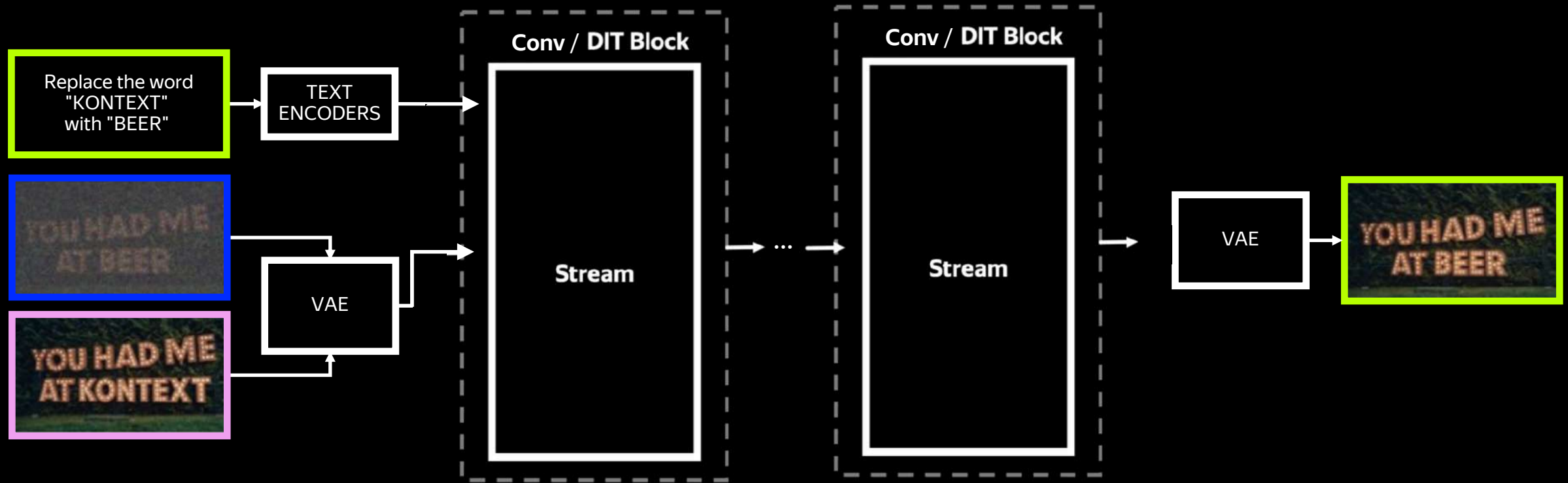


IP-Adapter

Semantic/Identity Conditioning

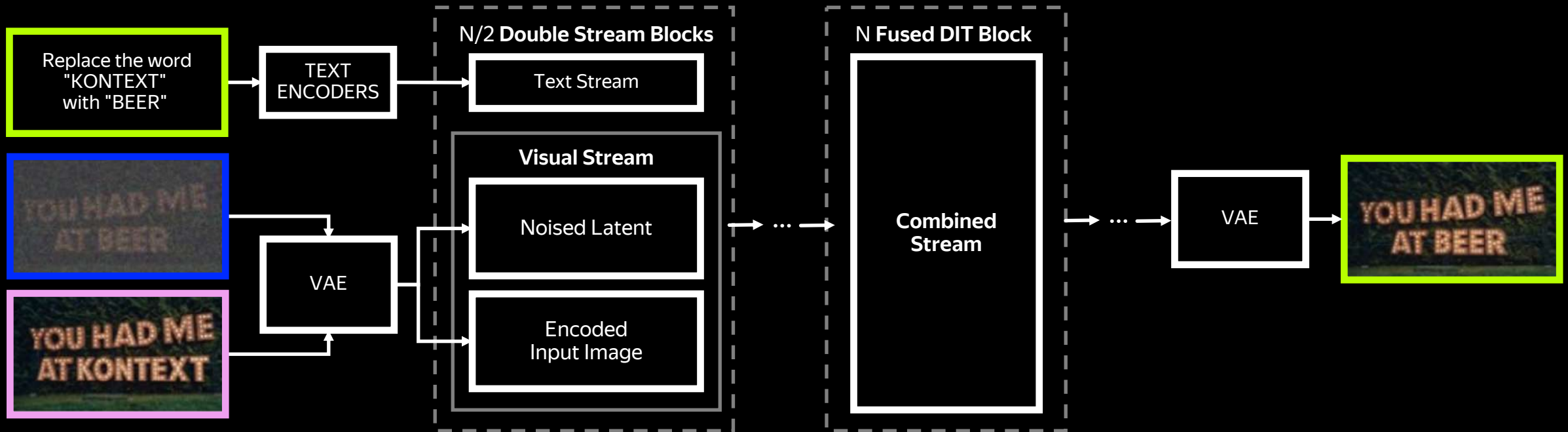


Transformers - just concat conditions



Transformers - the Flux way

Flux.1 Kontext



Classifier-Free Guidance

Training

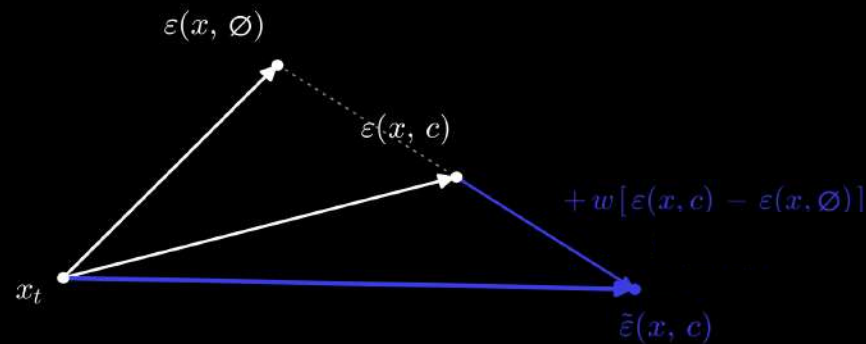
$$\mathcal{L} = \mathbb{E} \|\varepsilon_\theta(x, c) - \varepsilon\|^2 \quad c \leftarrow \emptyset \text{ with probability } p$$

Inference

$$\begin{aligned} \tilde{\varepsilon}(x, c) &= \varepsilon(x, \emptyset) + (1 + w) [\varepsilon(x, c) - \varepsilon(x, \emptyset)] \\ &= (1 + w) \cdot \varepsilon(x, c) - w \cdot \varepsilon(x, \emptyset) \end{aligned}$$

Implicit classifier

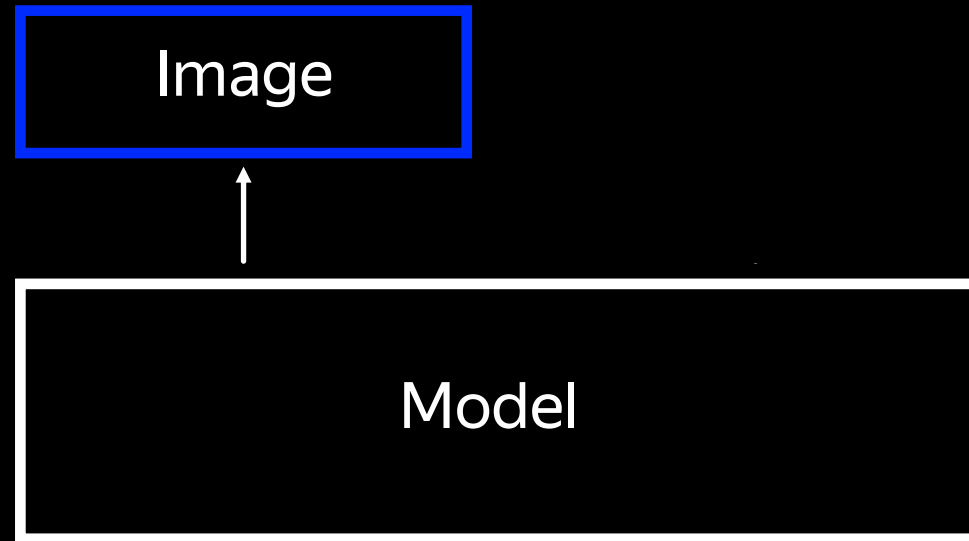
$$\nabla_{x_t} \log p(c \mid x_t) \approx - \frac{\epsilon_\theta(x_t, c, t) - \epsilon_\theta(x_t, \emptyset, t)}{\sigma_t}$$



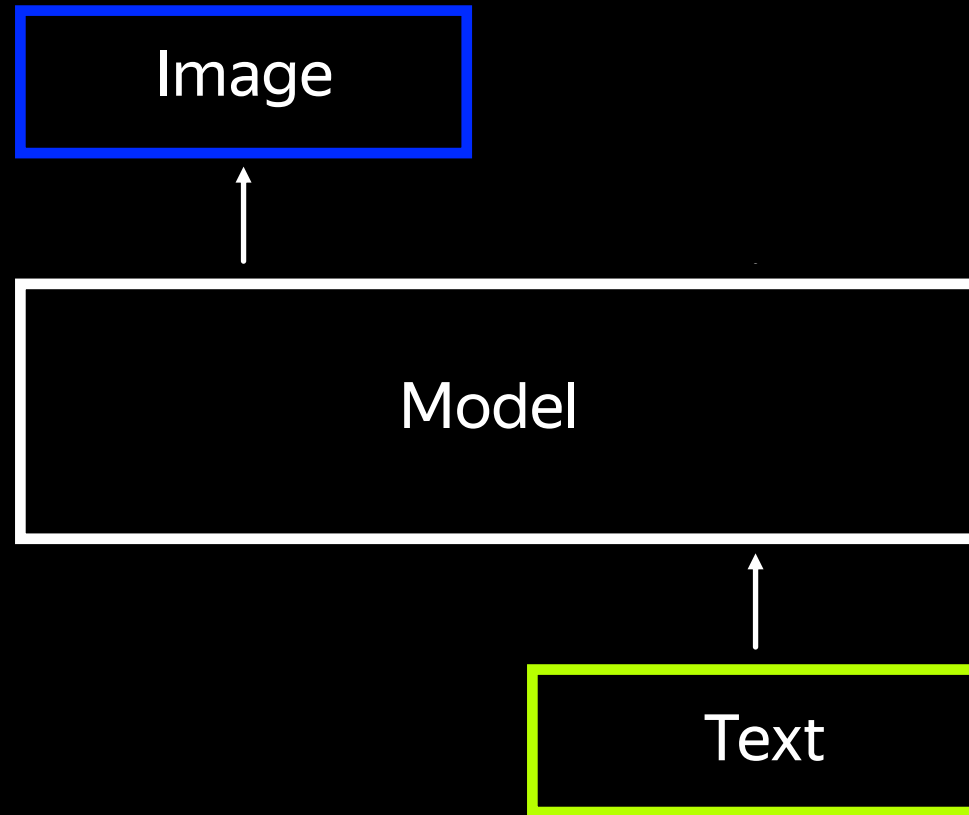
Extrapolate from $\varepsilon(x, \emptyset)$ through $\varepsilon(x, c)$ by w

$w = 0$ recovers $\tilde{\varepsilon}(x, c)$
 $w \nearrow$ samples from $p(x|c) \cdot p(c|x)^w \rightarrow$ fidelity $\uparrow$, diversity $\downarrow$
 $w \gg 1$ extrapolation leaves the data manifold

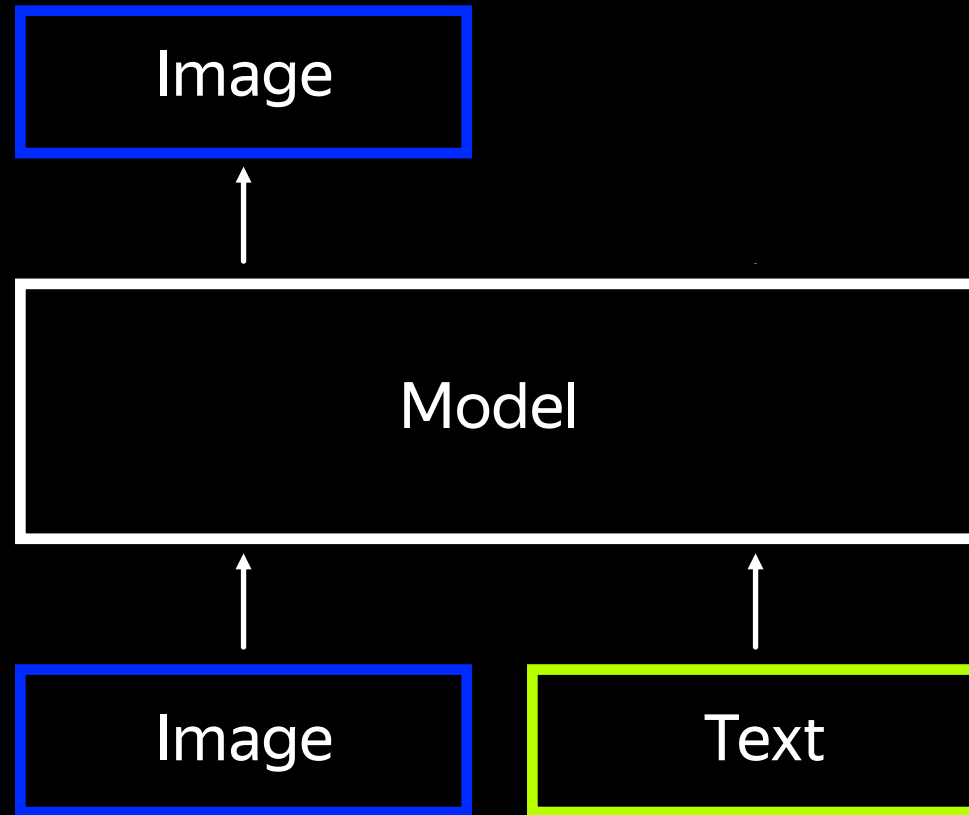
Recap



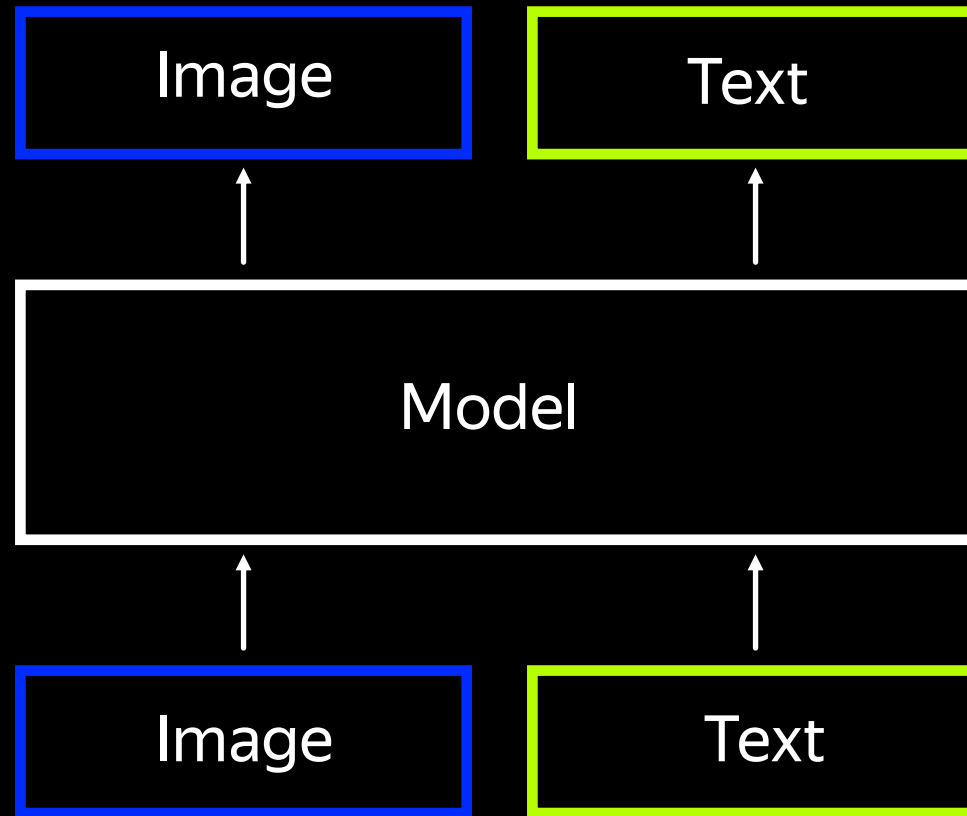
Recap



Recap



Multi-modal Generation



In reality we want multi-modal

Why?

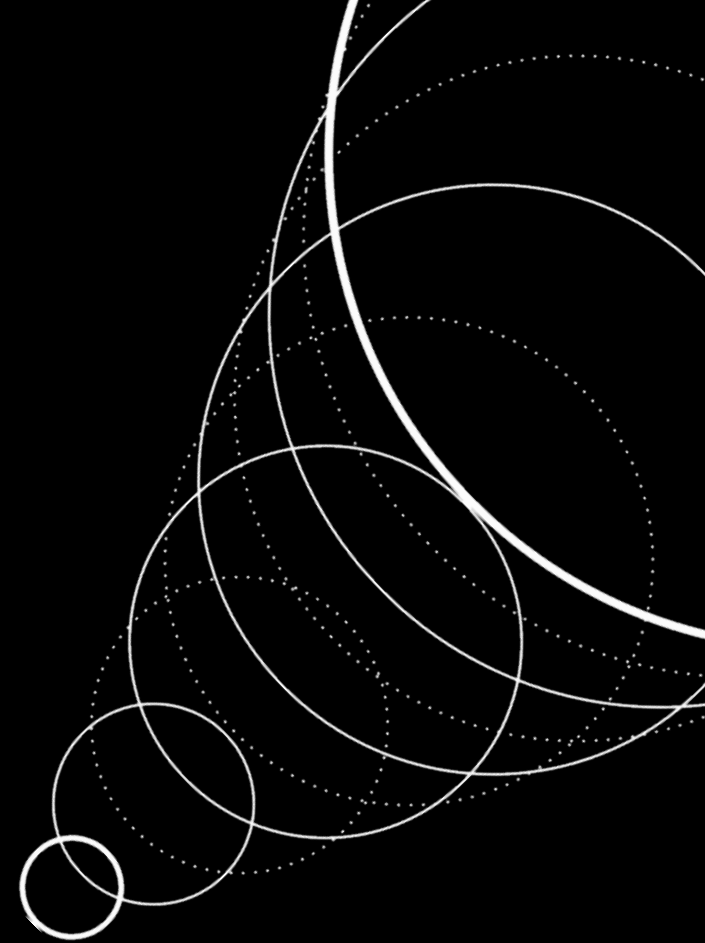
- 01 Bidirectional capability*
- 02 Interleaved / any-to-any generation
- 03 Emergent multimodal reasoning

* J. Zhang et. al., “Are Unified Vision-Language Models Necessary: Generalization Across Understanding and Generation”, 2025

The central design tension

Unify two main generative worlds

- Inherently continuous (visual): images, video, 3D, audio
- Inherently discrete (textual): text, code, math



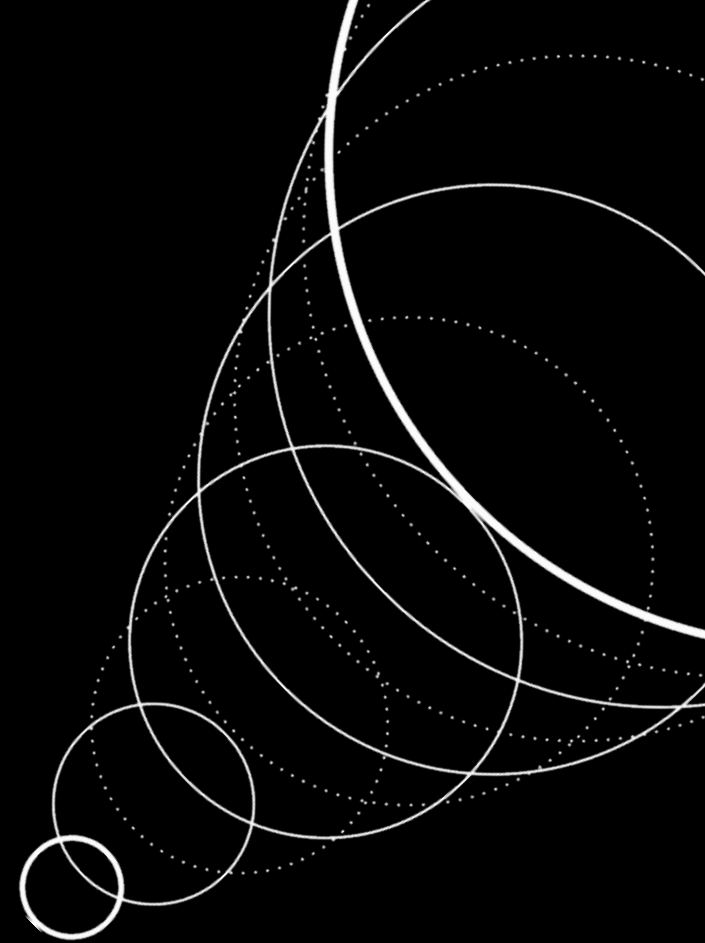
The central design tension

Unify two main generative worlds

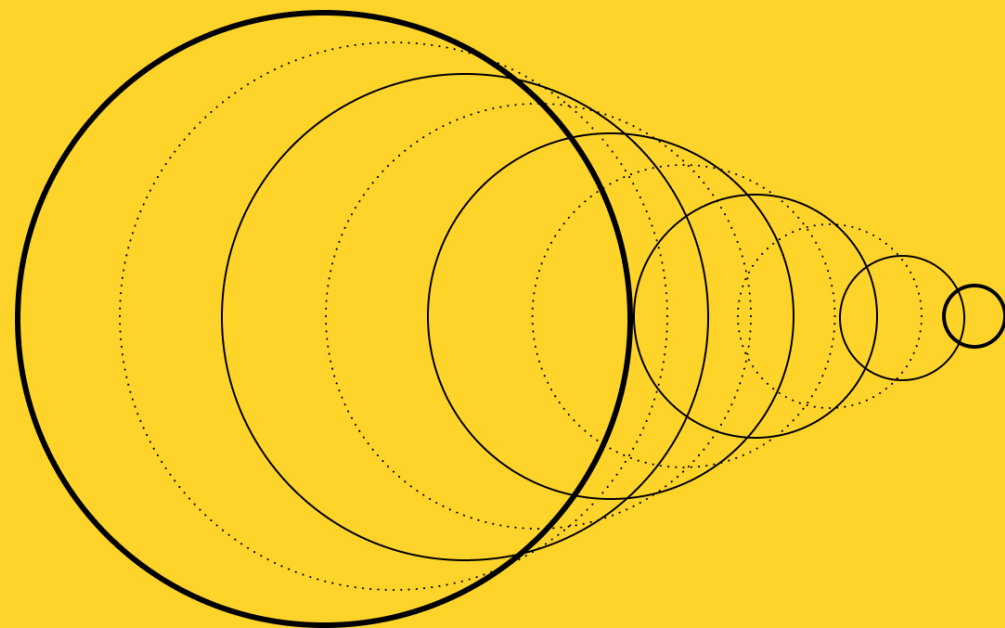
- Inherently continuous (visual): images, video, 3D, audio
- Inherently discrete (textual): text, code, math

Currently SOTA approaches

- discrete-to-discrete — LLM-like (e.g. aLLM)
- discrete-to-continuous — diffusion models
- mixed-to-discrete — VLM-like (e.g. AVLM)
- mixed-to-continuous — diffusion models

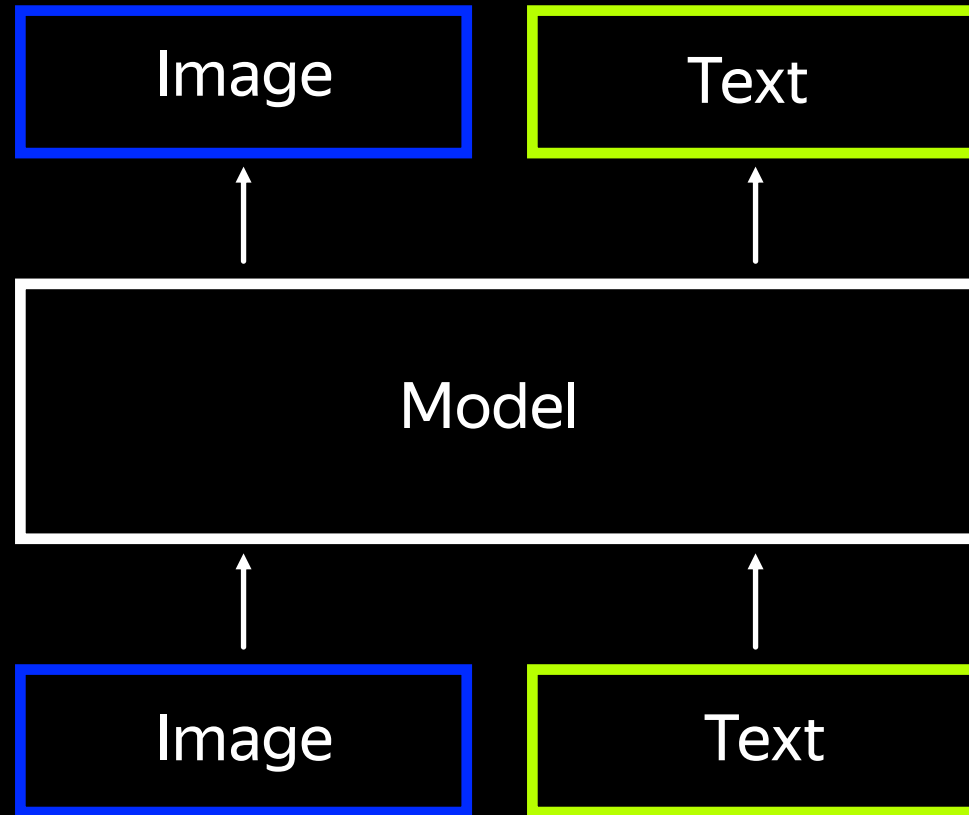


Yandex Research



**Towards Unified
Generative Models**

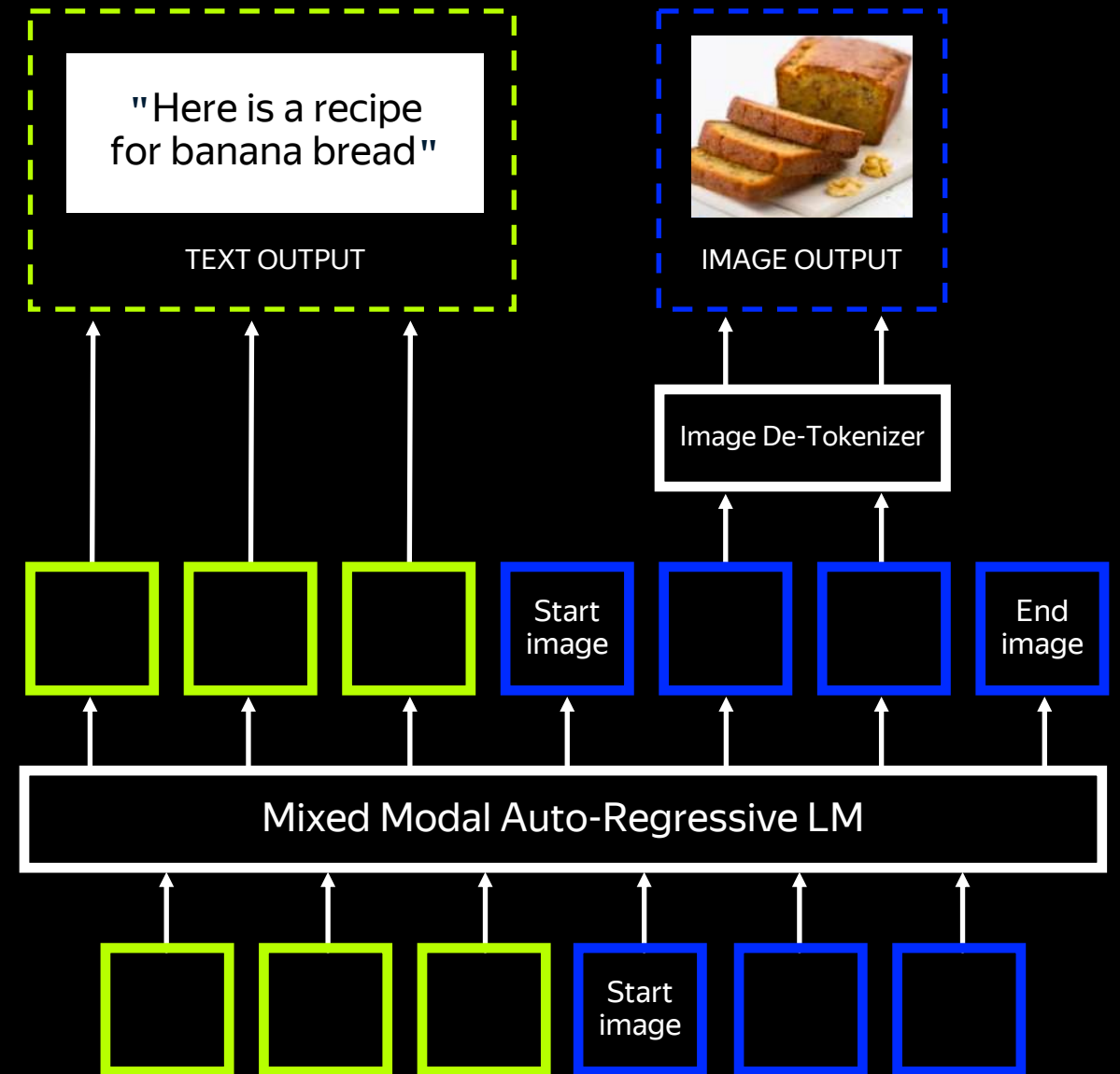
What kind of model would help us?



Single backbone

Single objective

Pure auto-regressive



Single backbone

Single objective

Pure auto-regressive



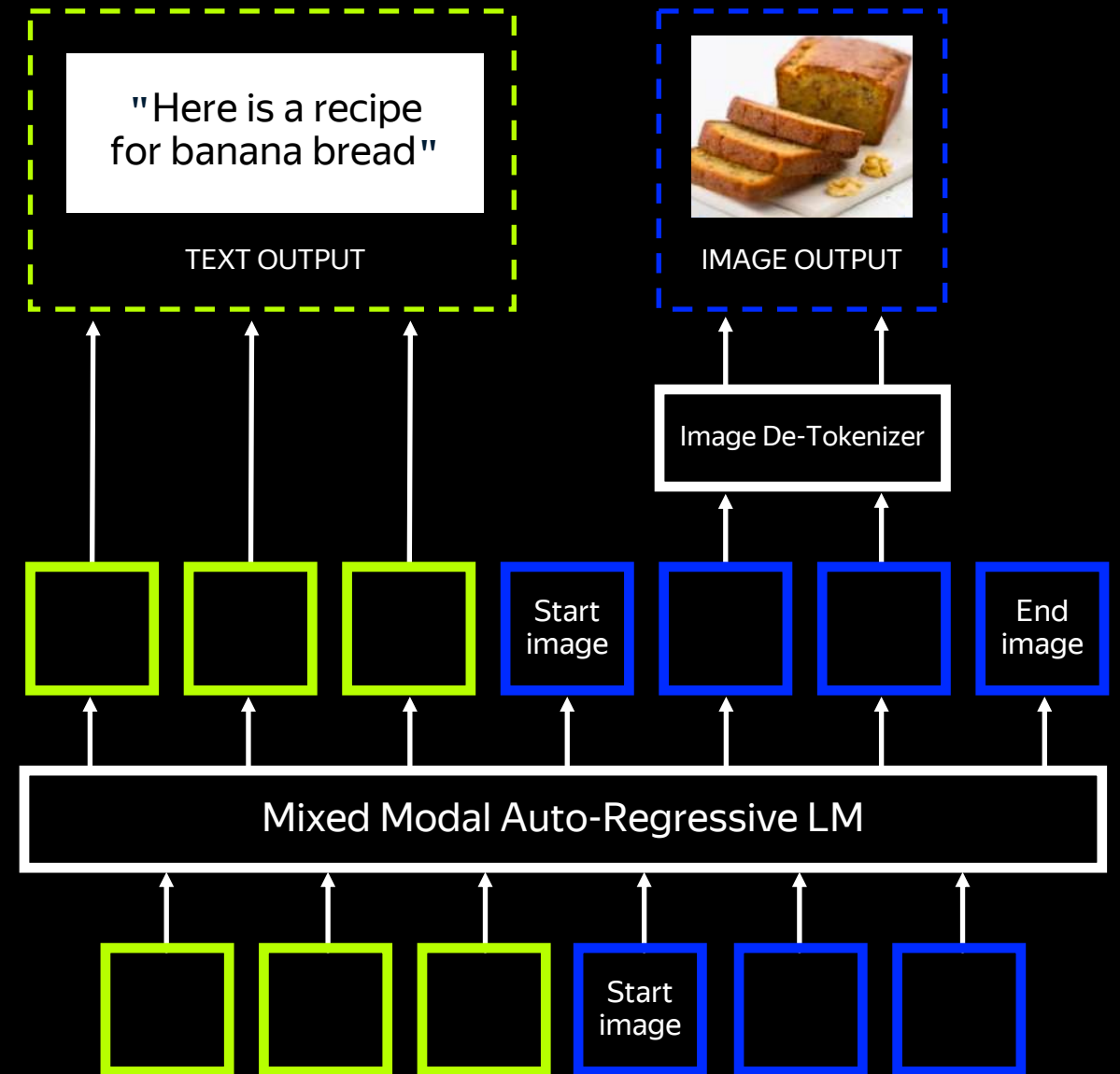
Uniform implementation



Raster order is not native to images



Discretisation ruins image quality



Single backbone

Single objective

Pure auto-regressive



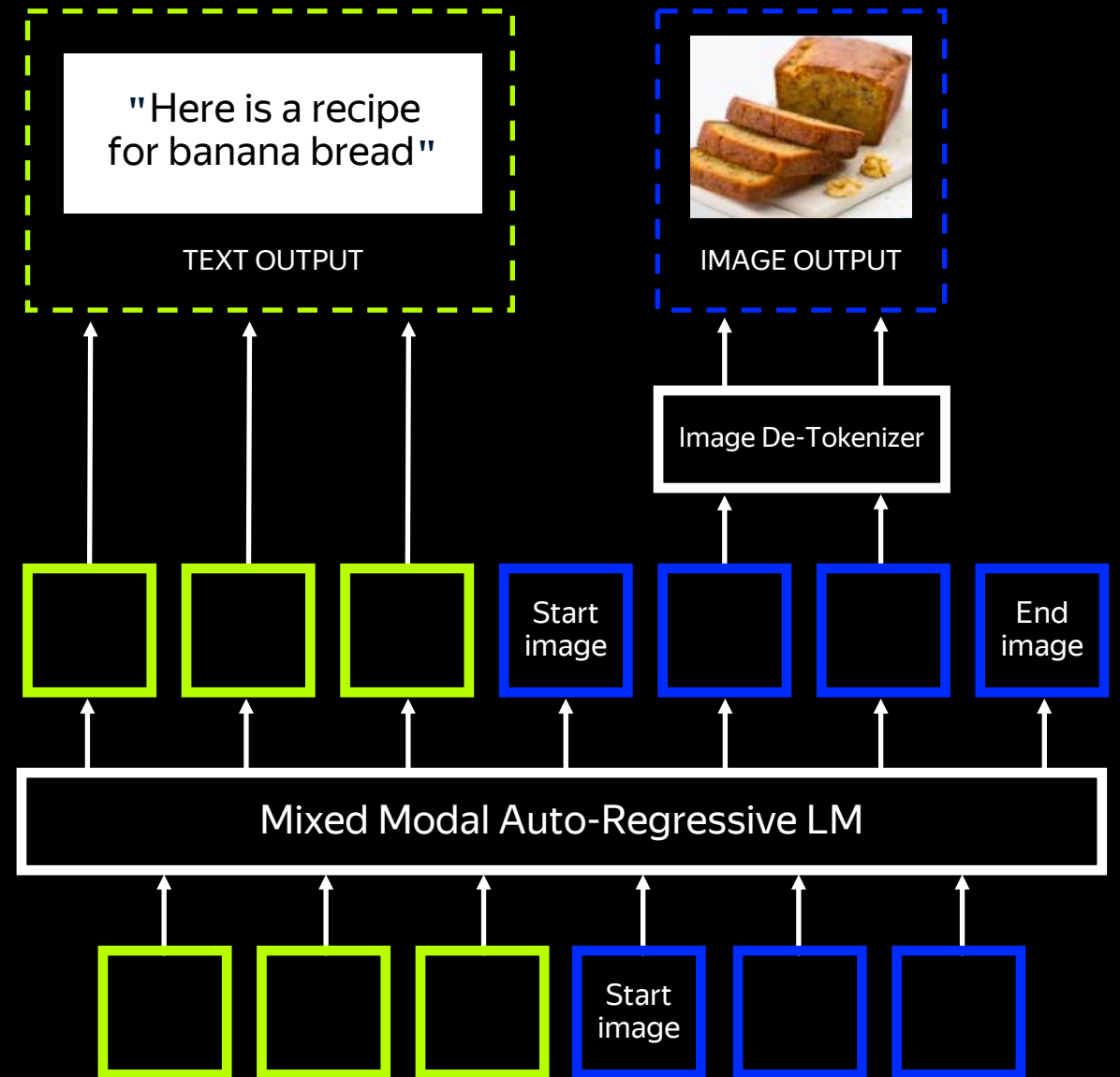
Uniform implementation

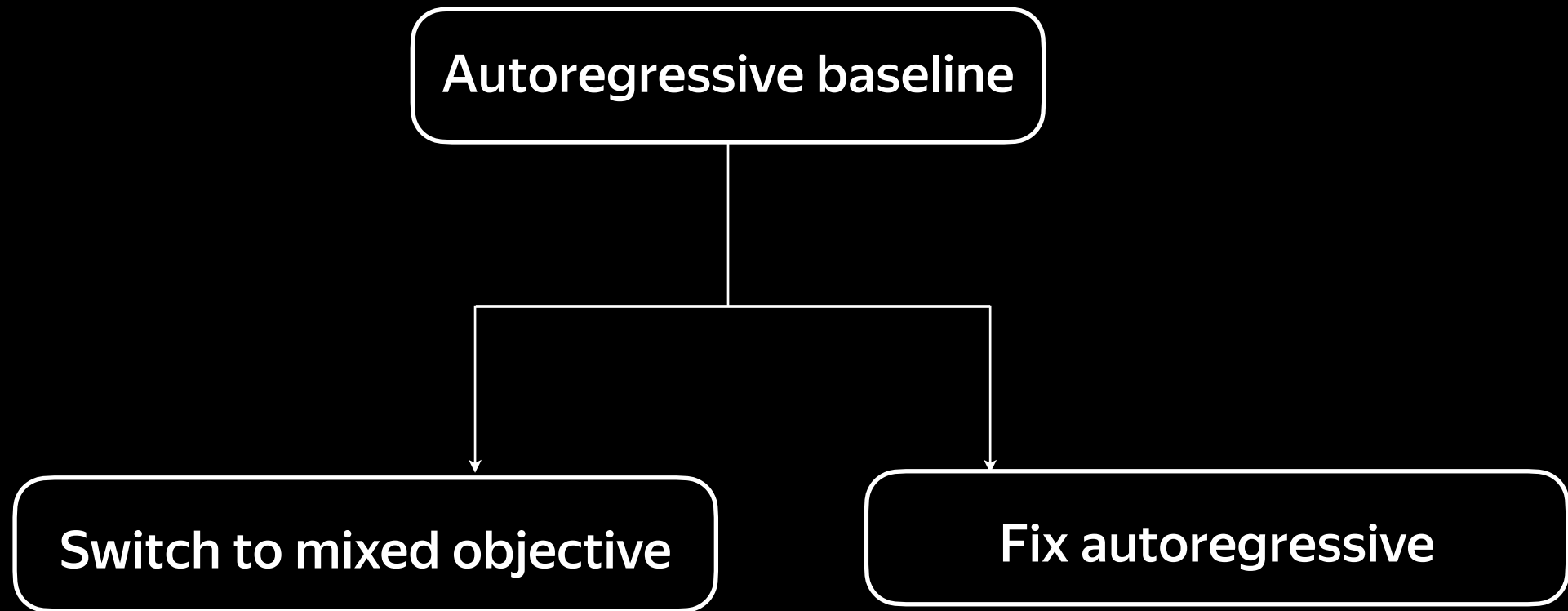


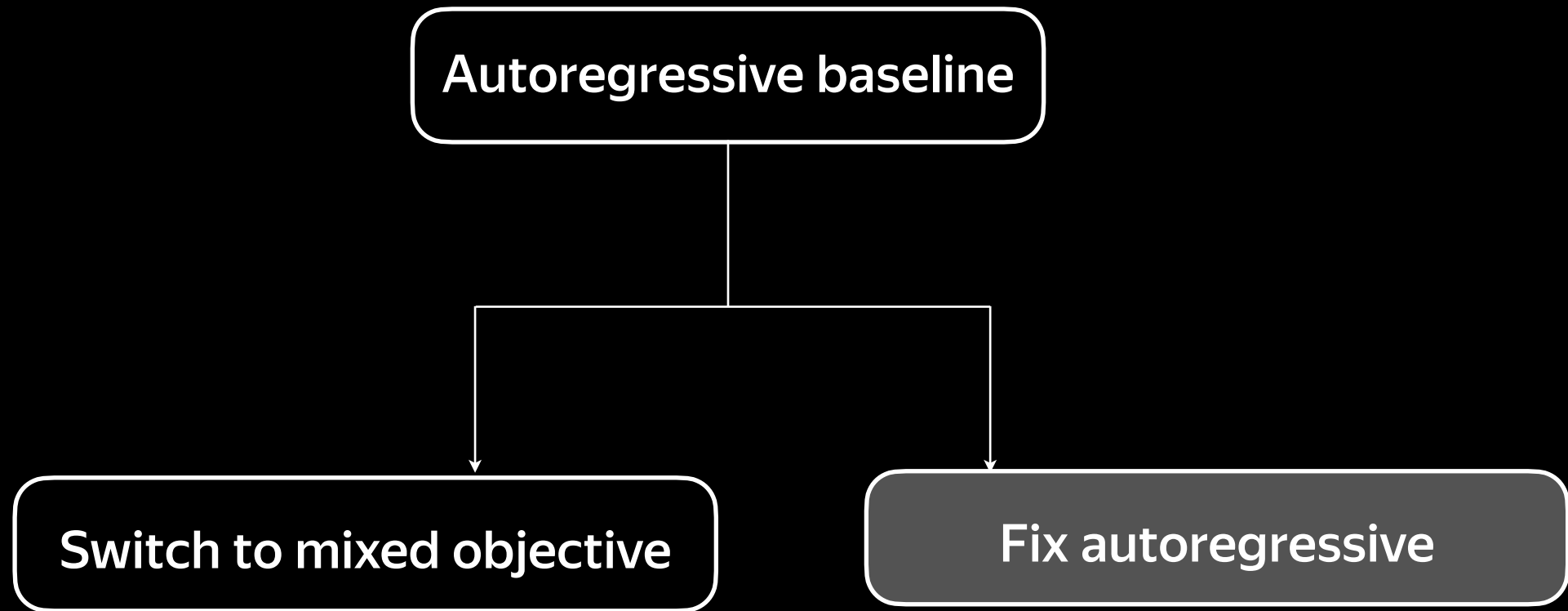
Raster order is not native to images



Discretisation ruins image quality



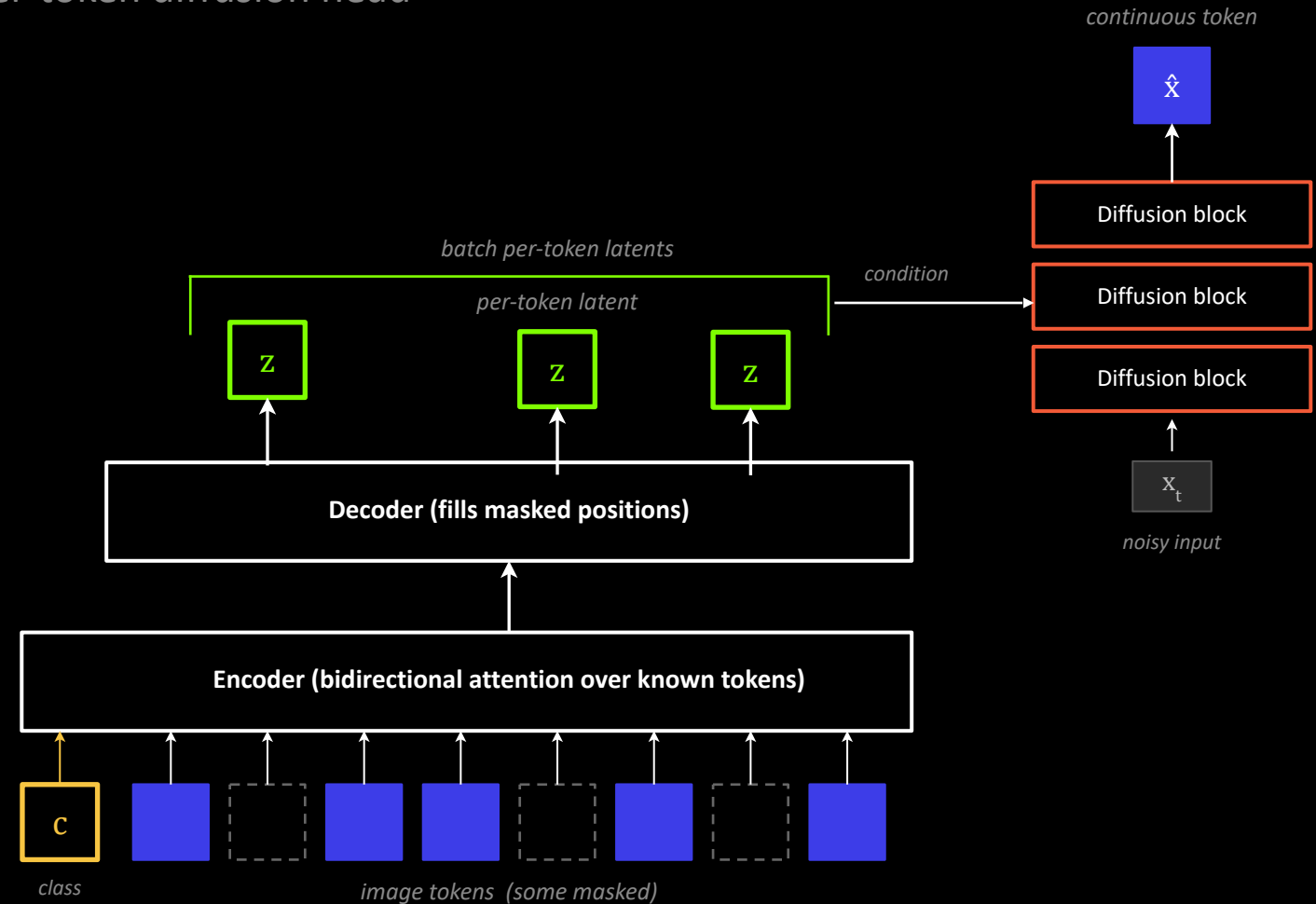




MAR: AR without Vector Quantization

Autoregression over continuous tokens via a per-token diffusion head

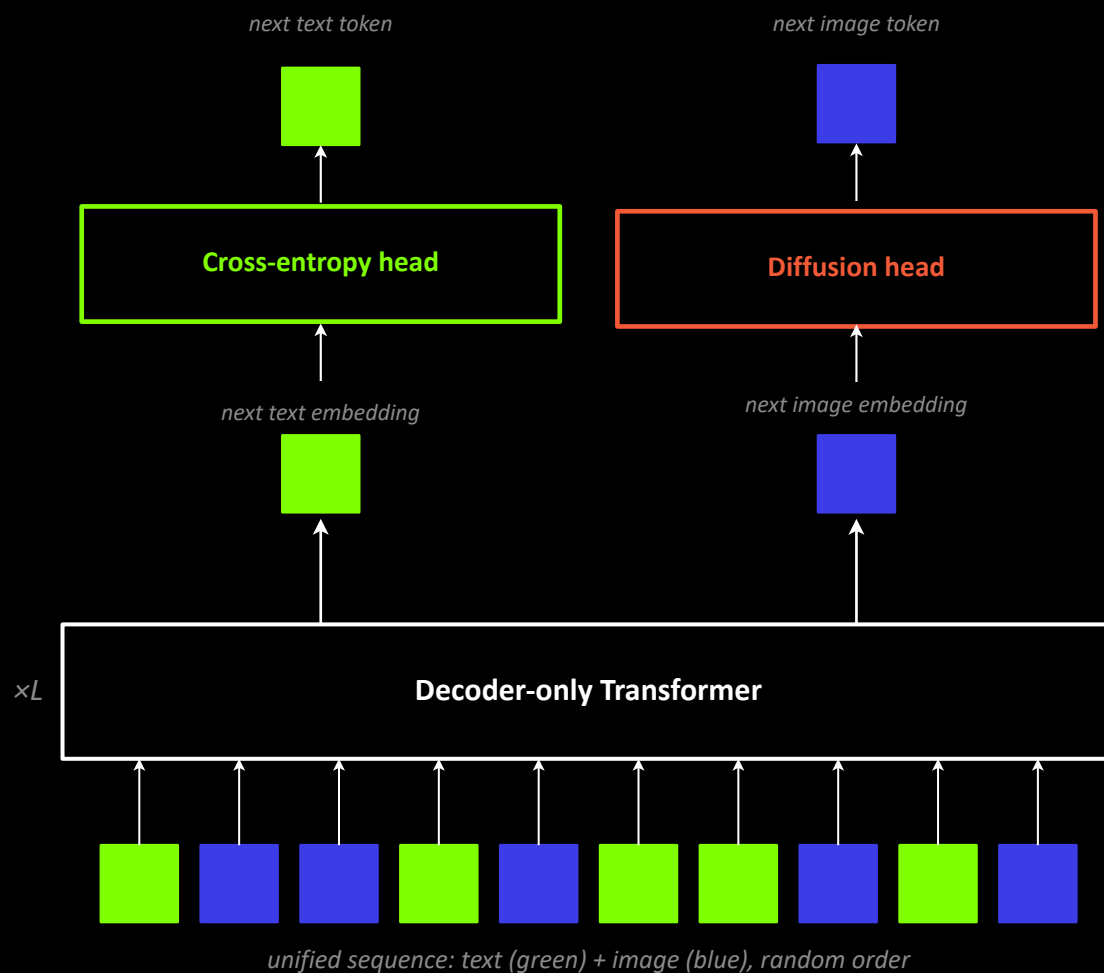
- + Continuous tokens via small diffusion head
- + Random-order generation, many tokens per step
- + Bidirectional attention inside the image
- Class-conditioned only (no text prompts yet)



Fluid | UniFluid

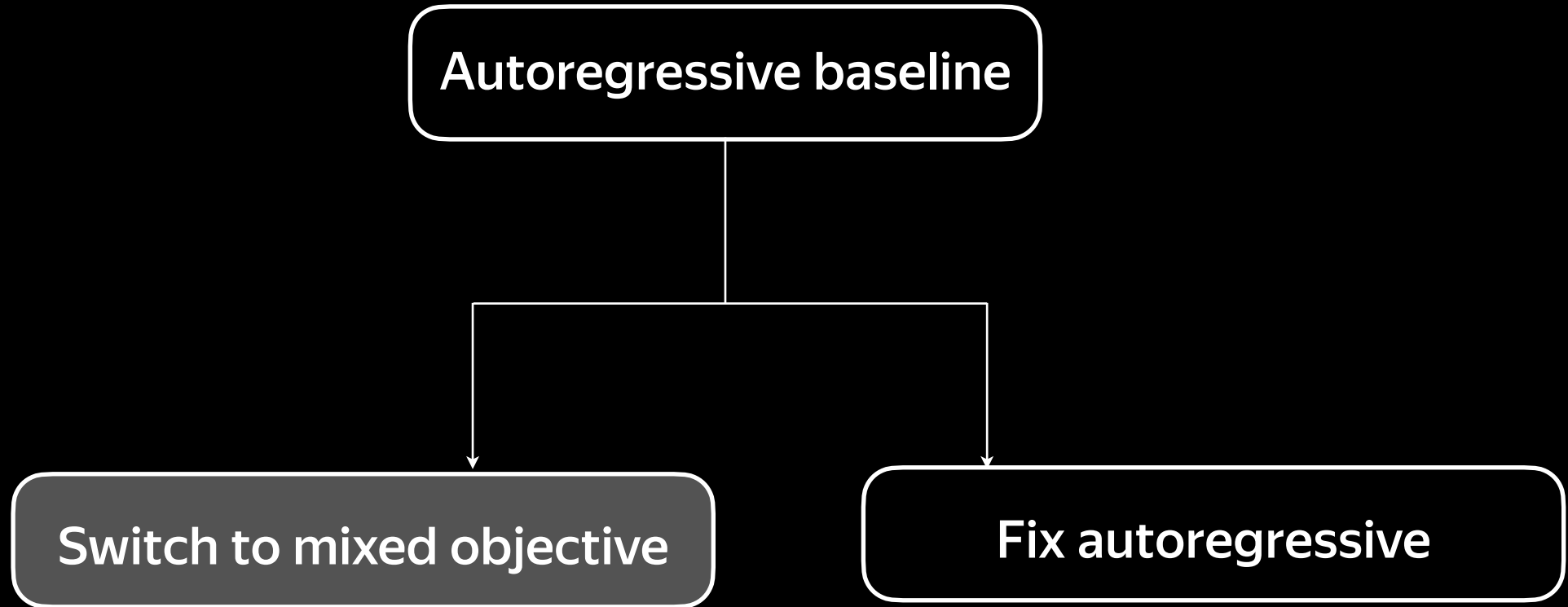
Scaling the MAR paradigm to text-to-image

- +** Fluid: random-order AR on continuous tokens, scales to 10.5B
- +** UniFluid: one decoder-only transformer for gen & understanding
- +** Two heads: diffusion head for images, CE head for text
- Tasks trade off — loss balance weight must be tuned carefully



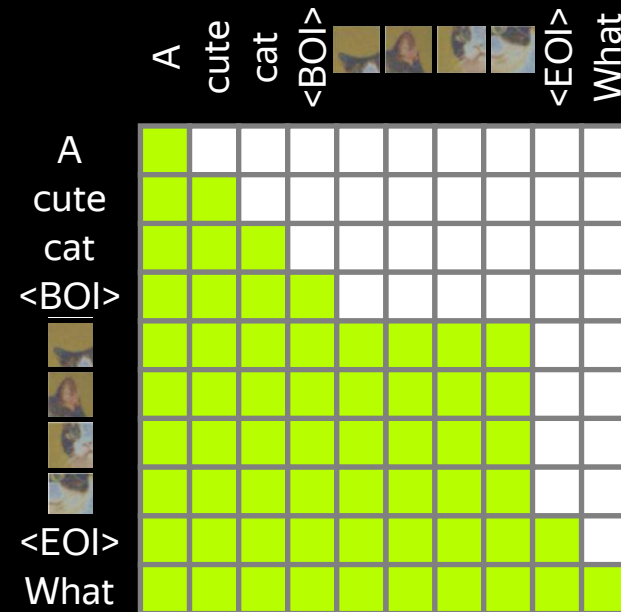
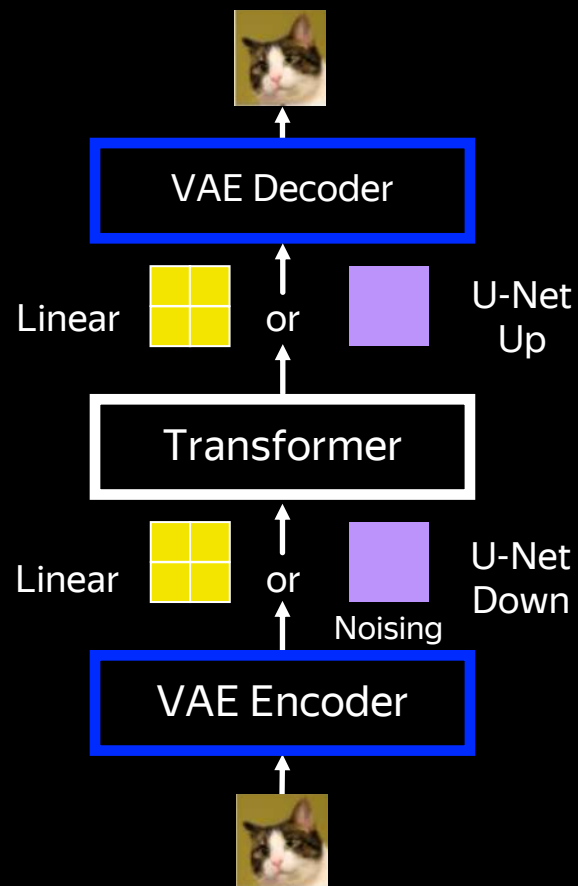
$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda \cdot \mathcal{L}_{\text{diff}}$$

joint next-token loss — balance weight λ matters



Single backbone Mixed objective

Transfusion



$$\mathcal{L}_{Transfusion} = \mathcal{L}_{LM} + \lambda \times \mathcal{L}_{DDPM}$$

Single backbone Mixed objective

Transfusion



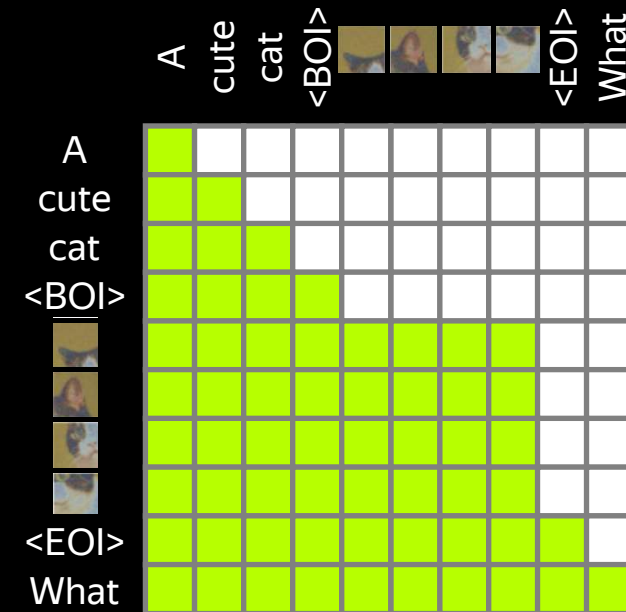
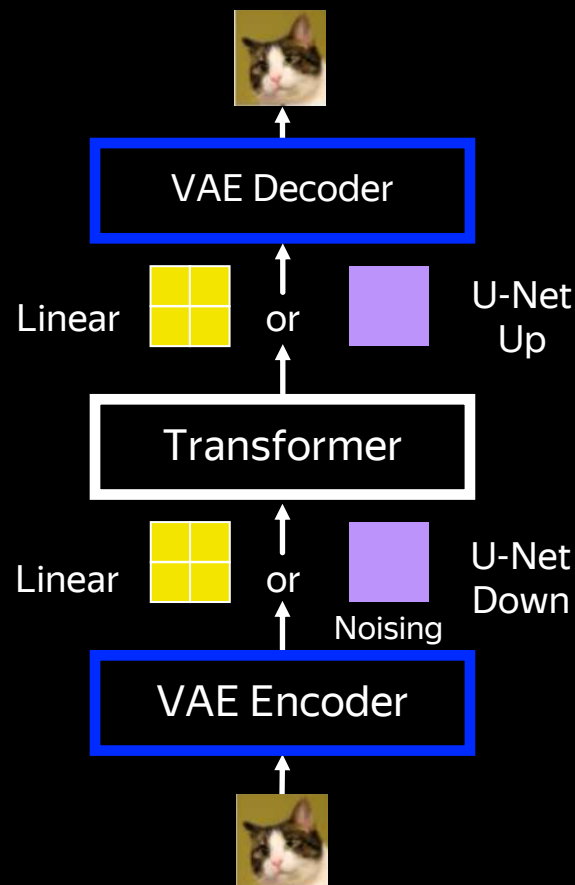
Unified representation
in a single transformer



Causal attn for text,
bidirectional for images



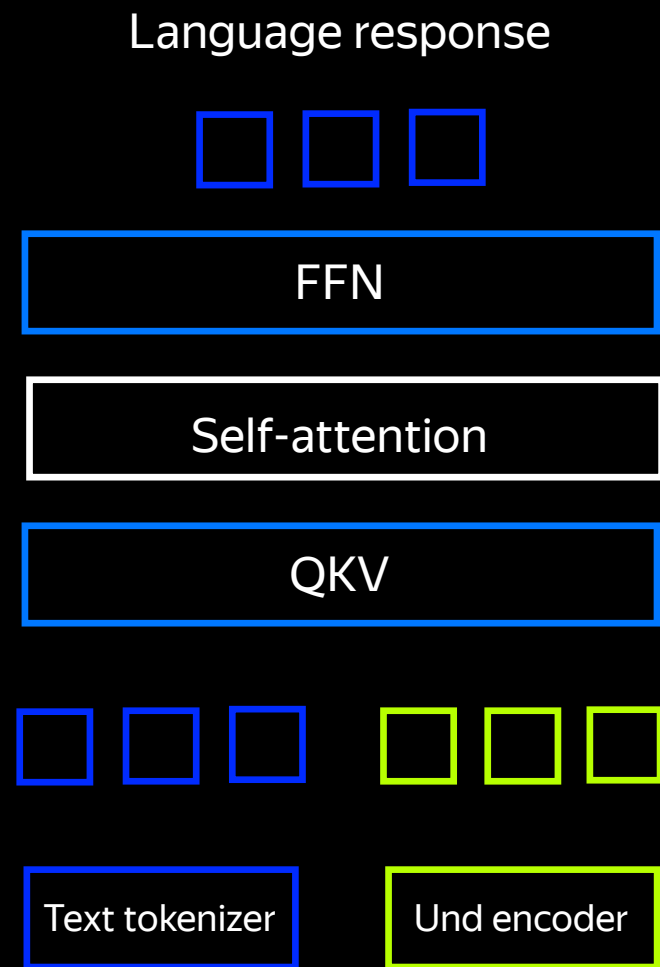
Continuous encoding
of images



$$\mathcal{L}_{Transfusion} = \mathcal{L}_{LM} + \lambda \times \mathcal{L}_{DDPM}$$

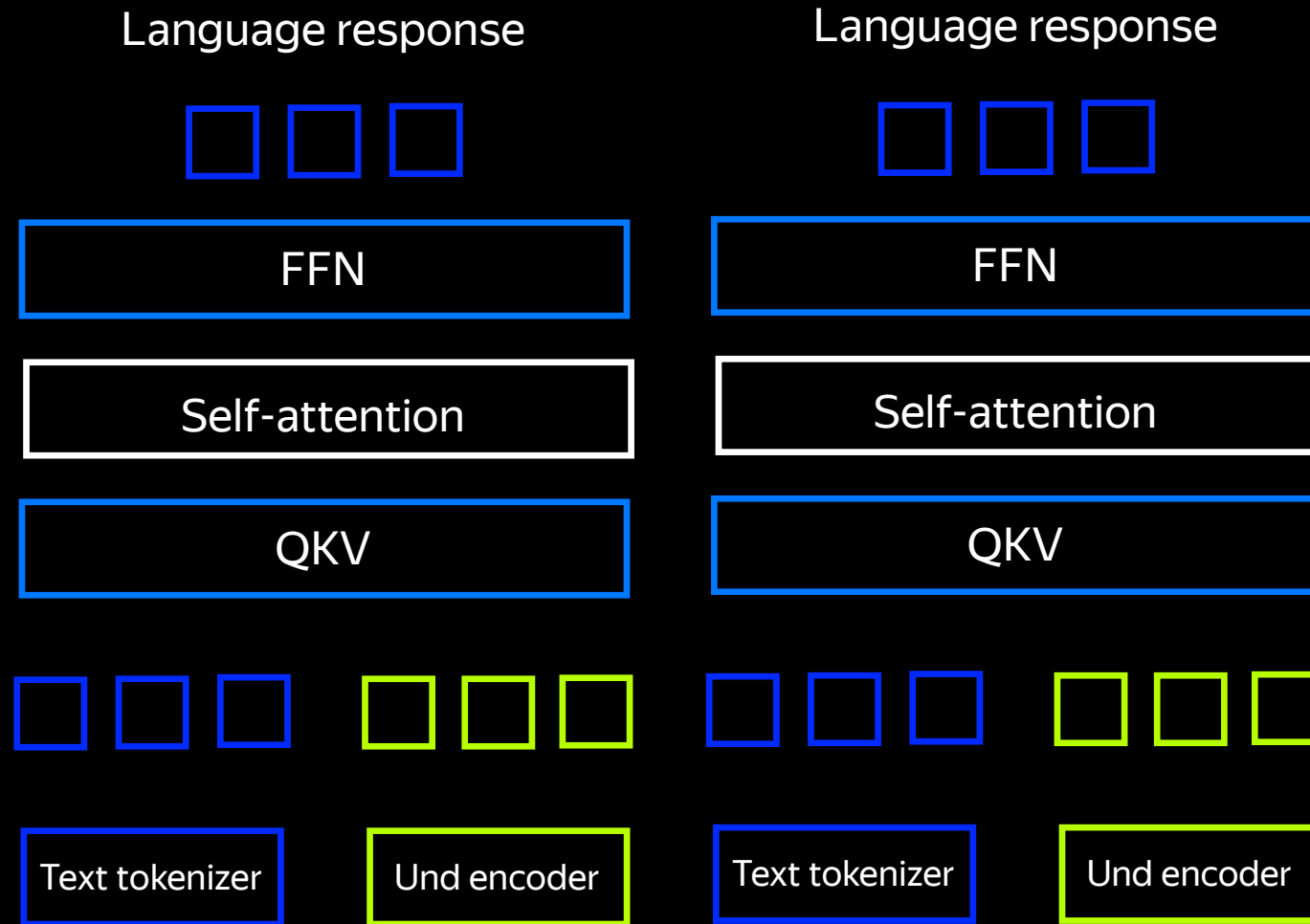
Two backbones - Mixed objective

BAGEL



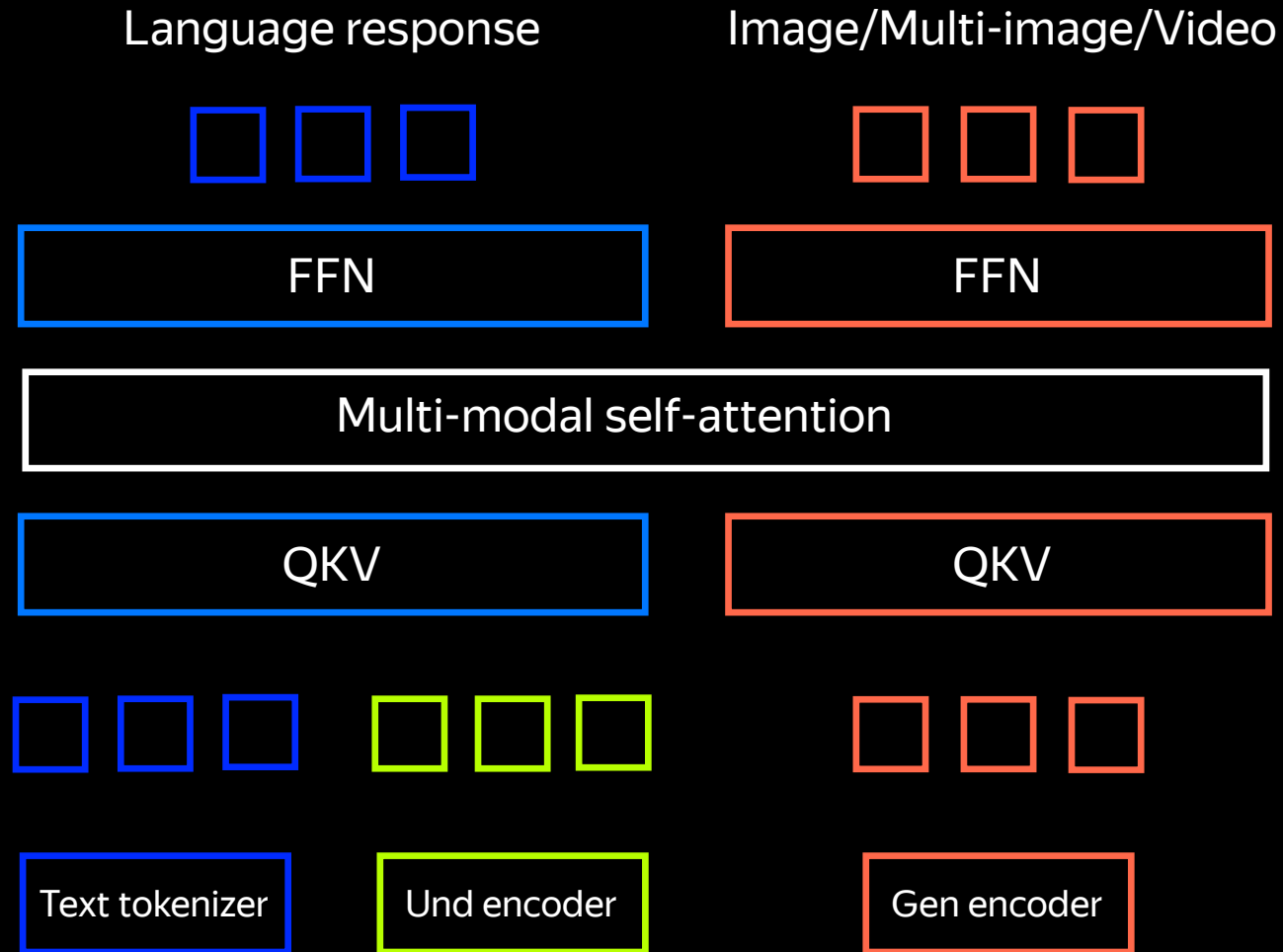
Two backbones - Mixed objective

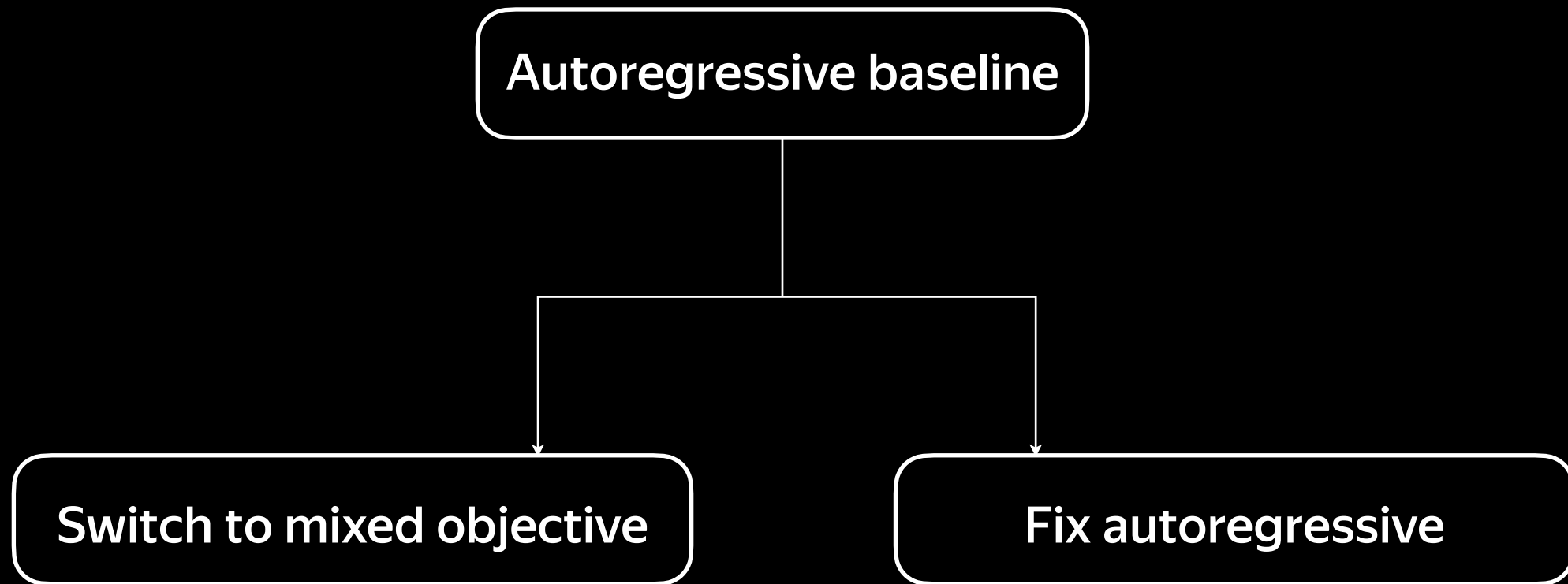
BAGEL

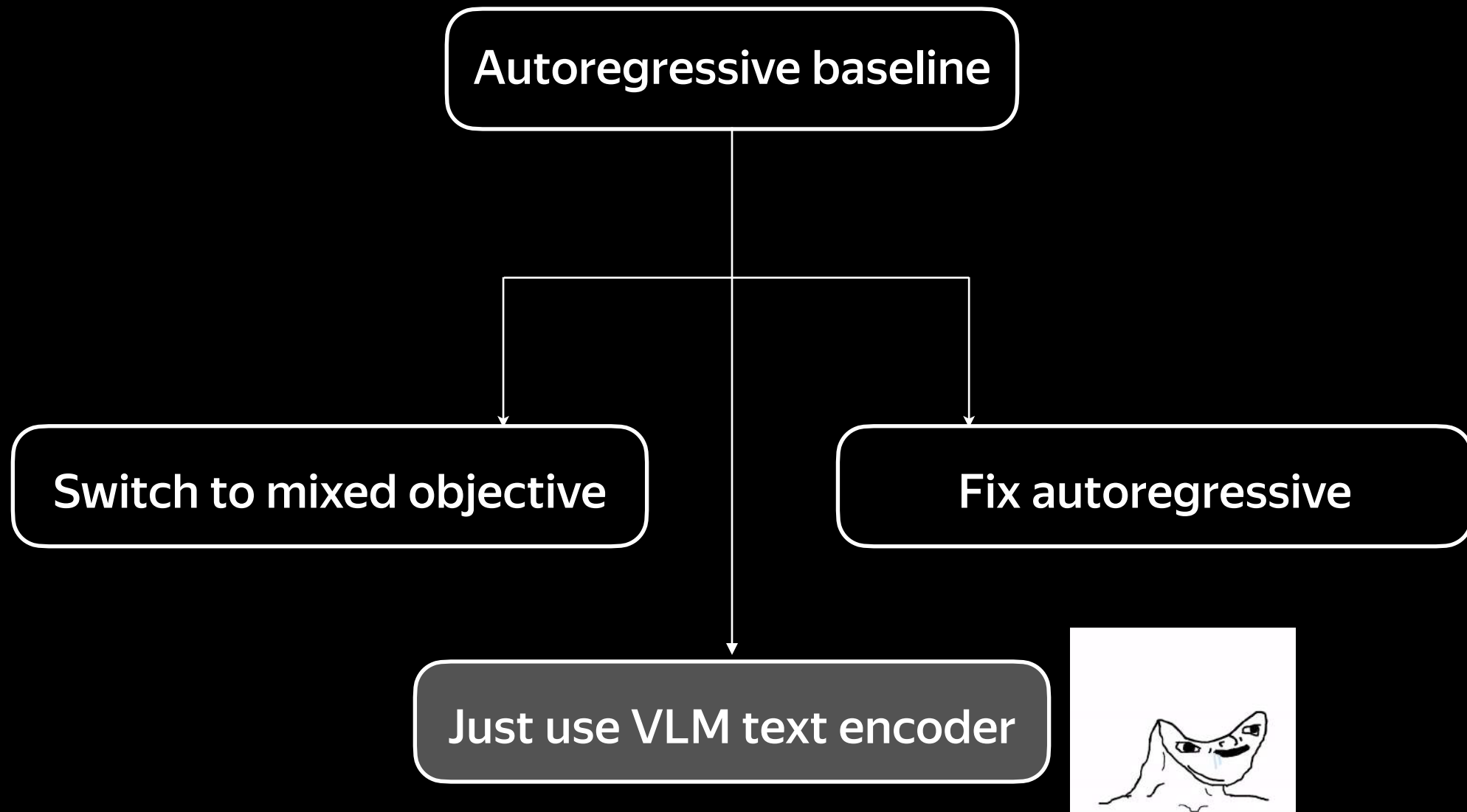


Two backbones - Mixed objective

BAGEL







NexusGen | Qwen Image



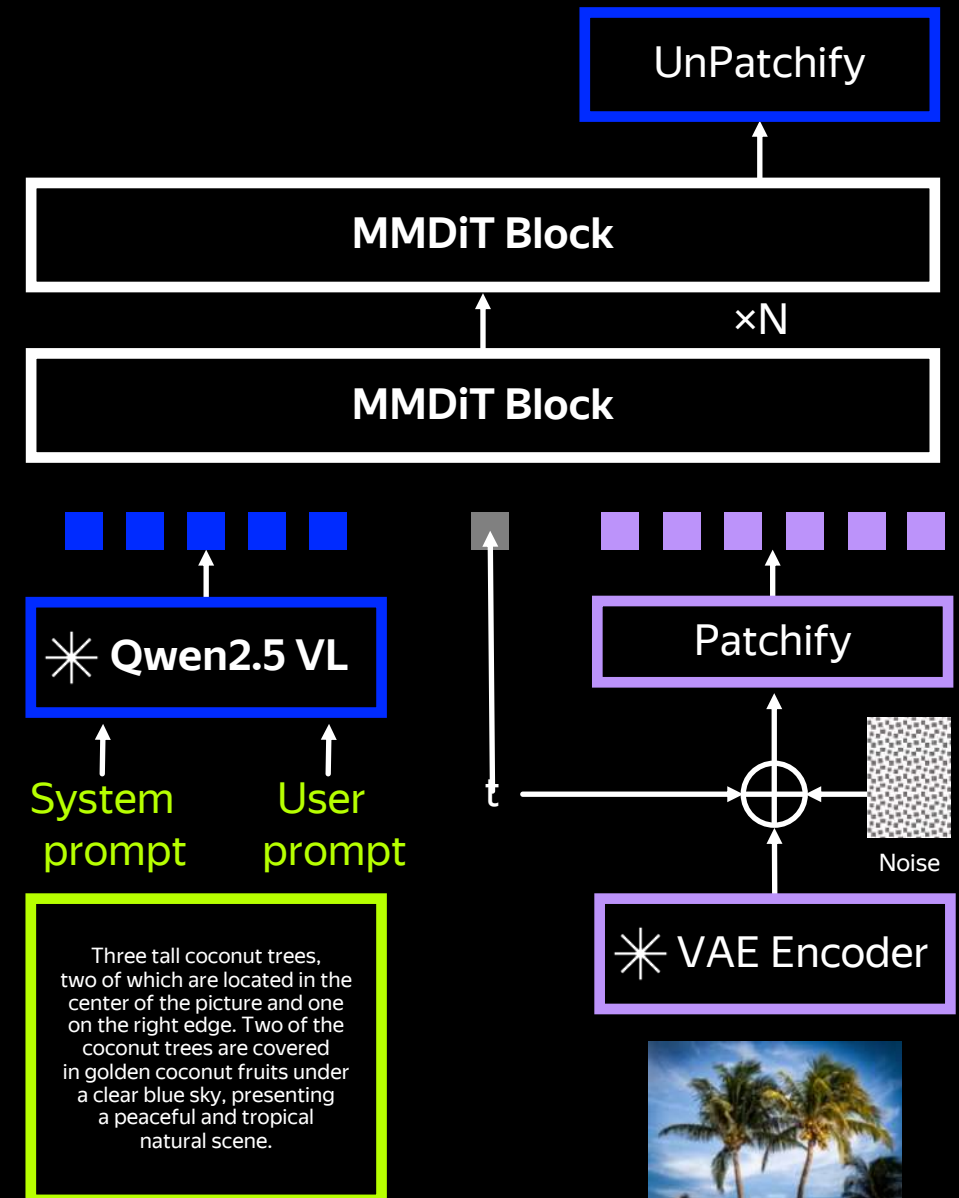
Re-use pre-trained diffusion denoiser

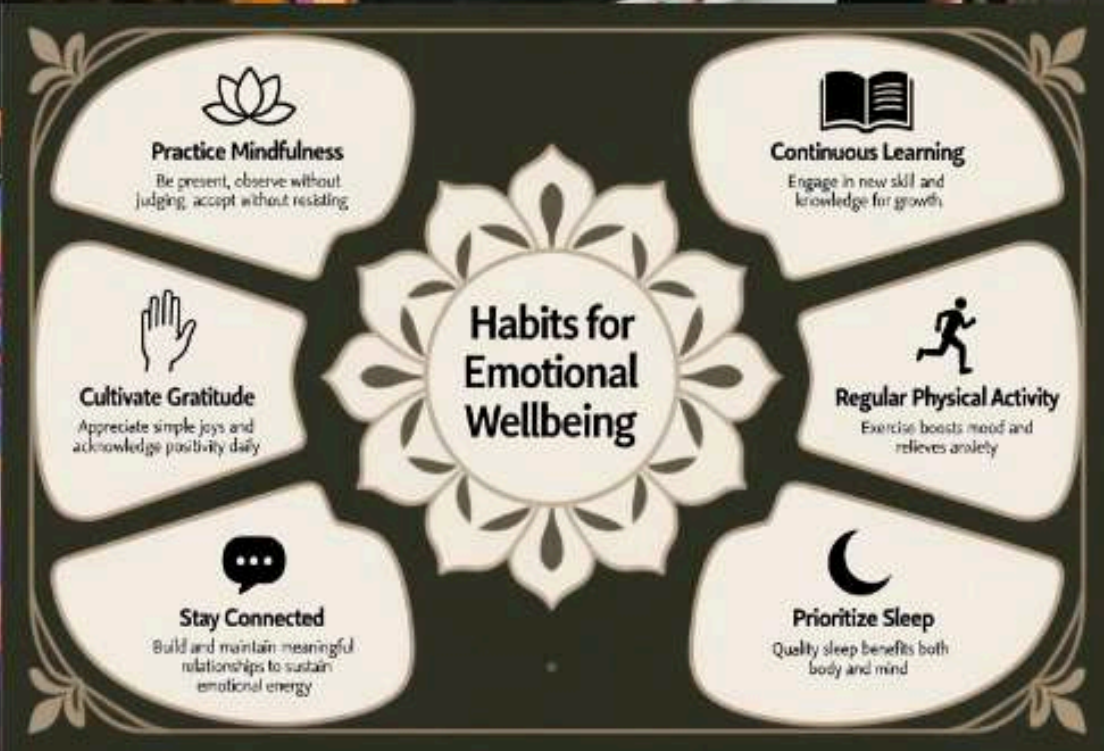
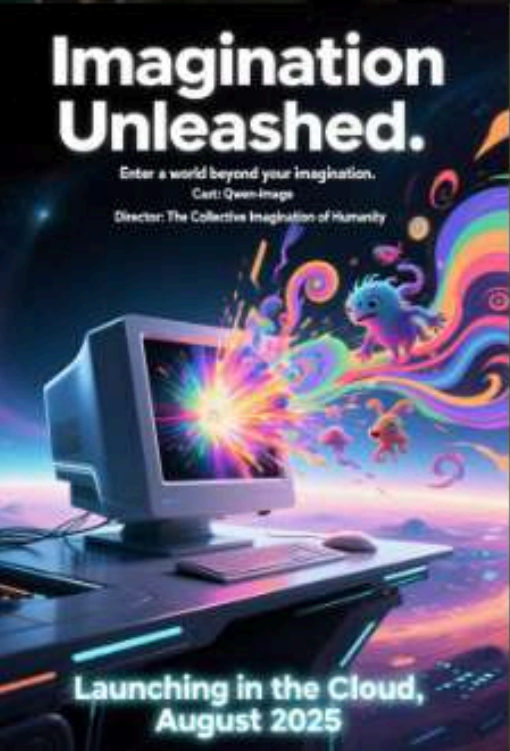


Should be cheaper to train, right?*



Hard to align with good quality





How hard is it to train the connector?

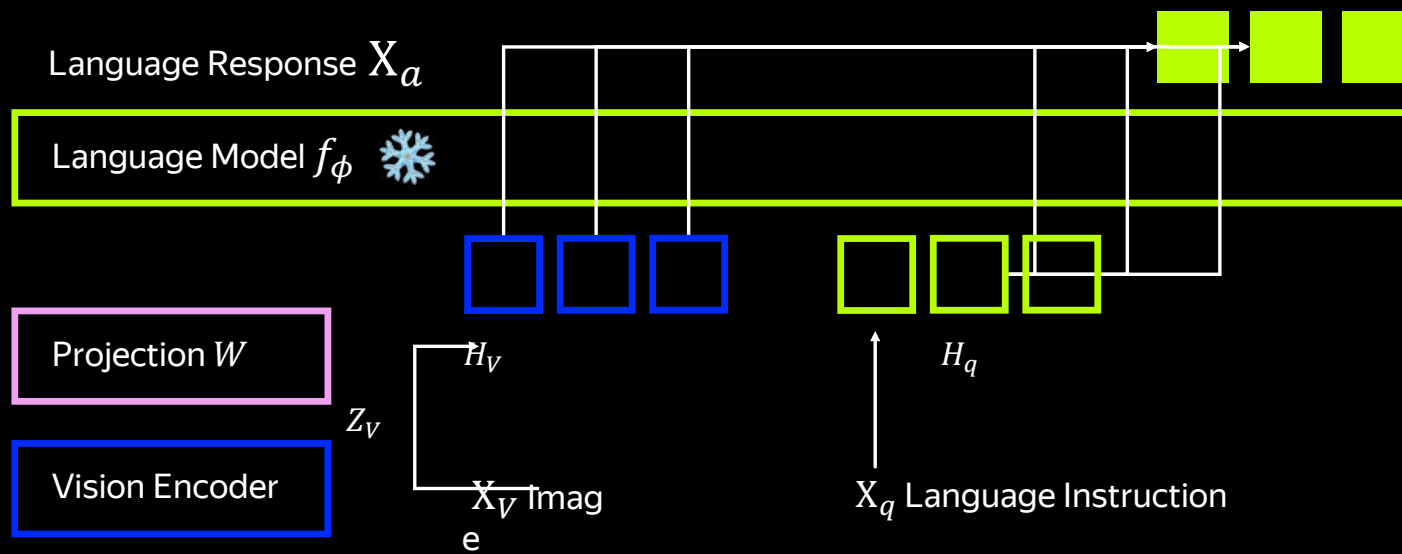
“a pair of cherries, dressed in a delicate ballerina outfit”



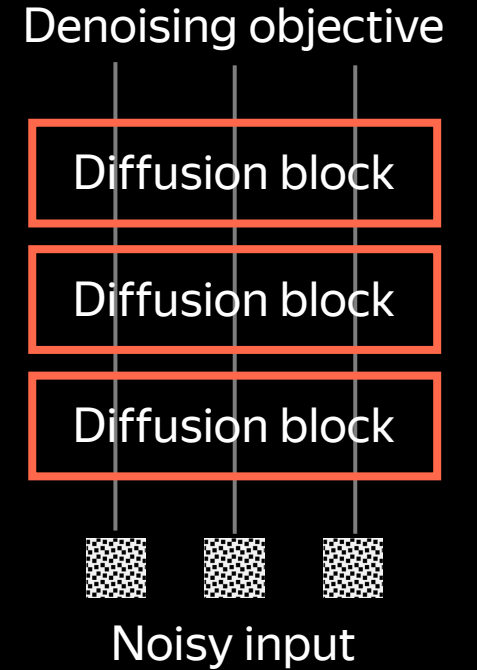
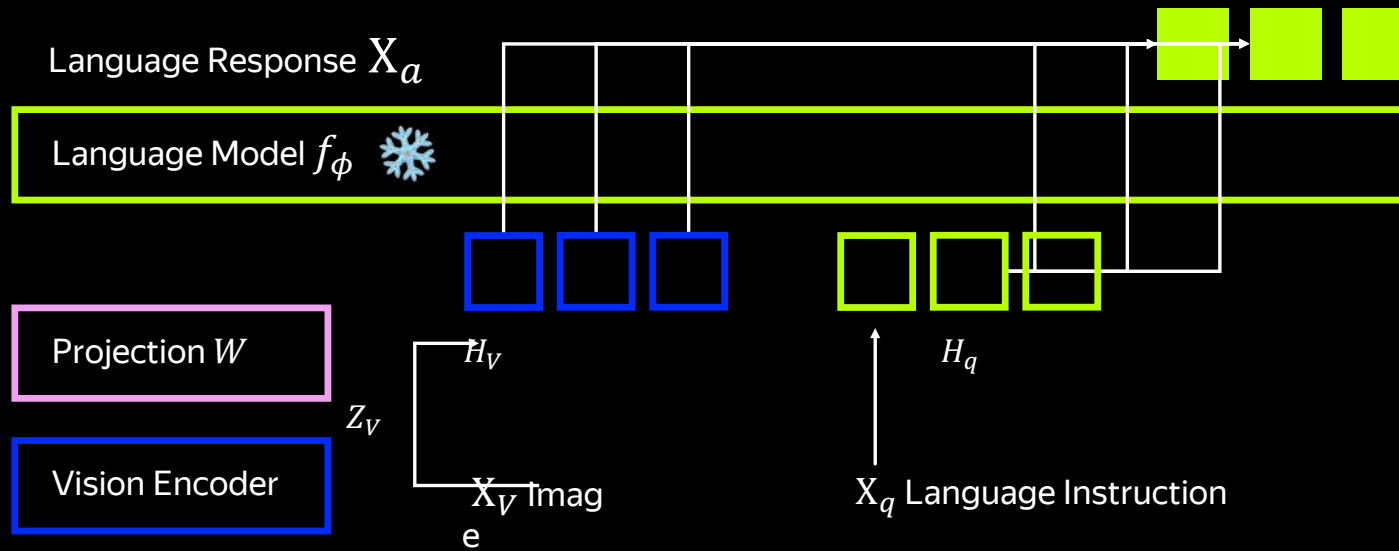
- Images generated with **NexusGen** have low quality
- Did not train from scratch
- Images generated with **Qwen Image** are much better
- Trained from scratch

* Very much a hand-waving argumentation, lots of other variables are not aligned between setups

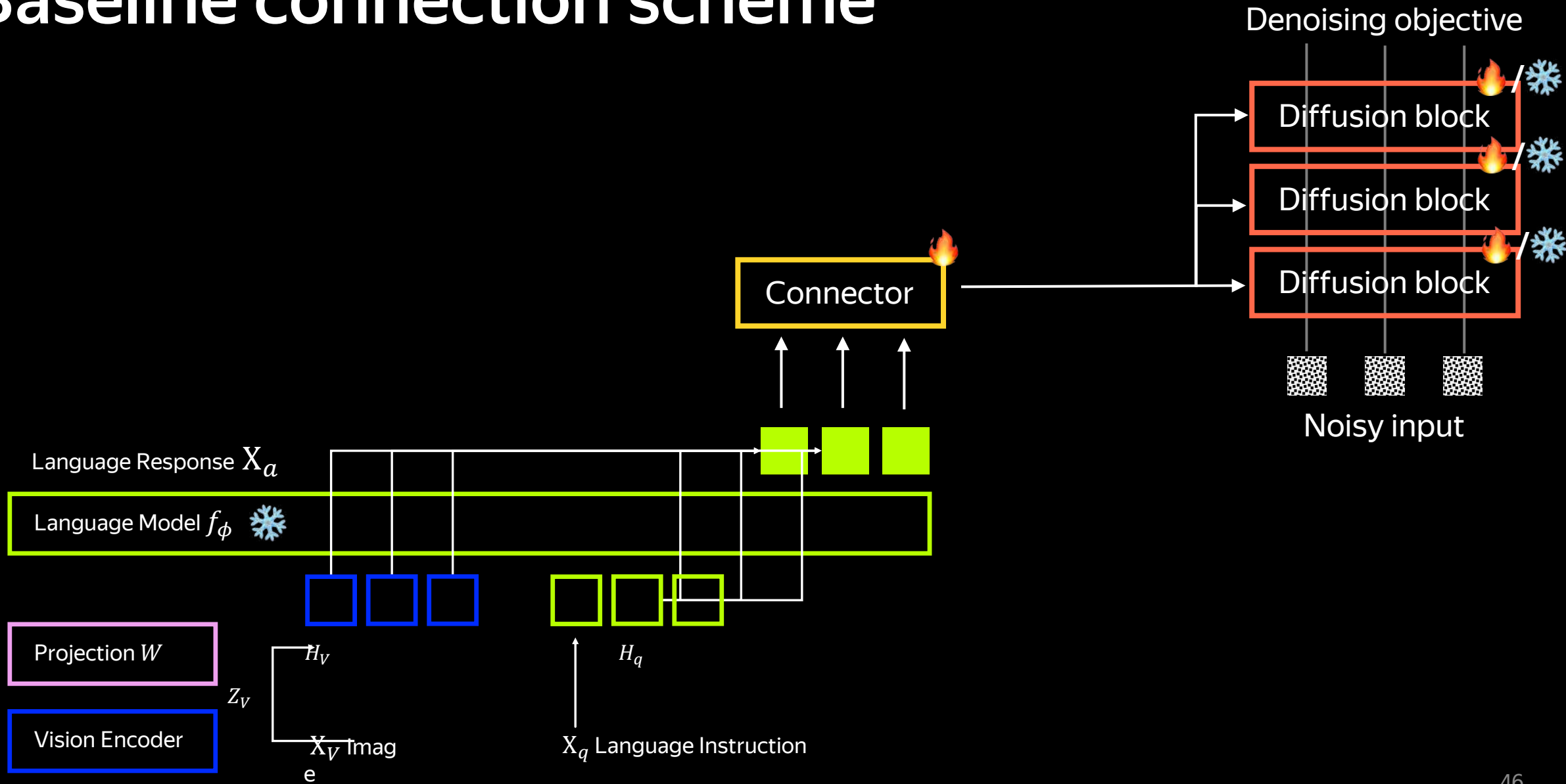
Baseline connection scheme



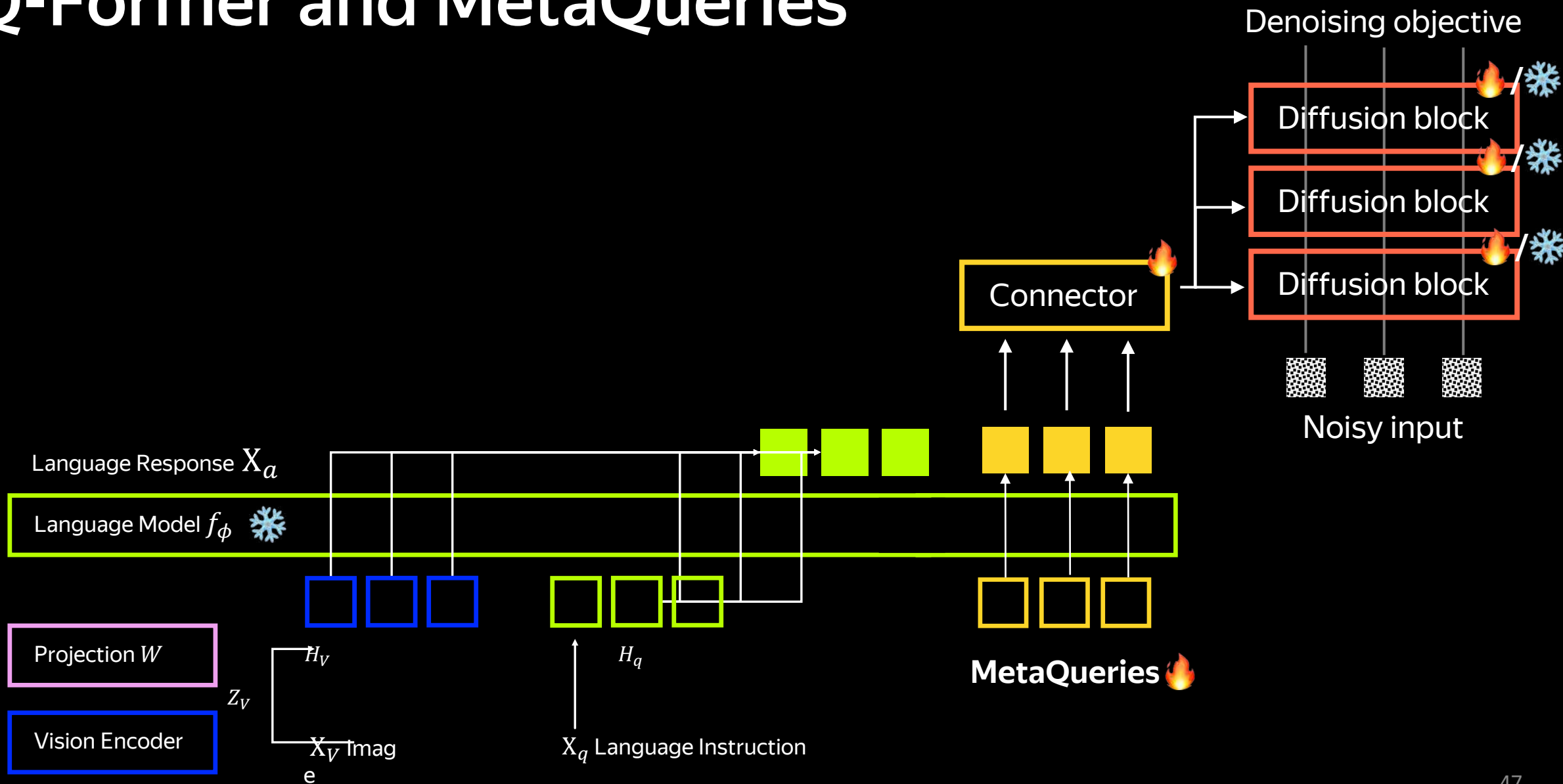
Baseline connection scheme



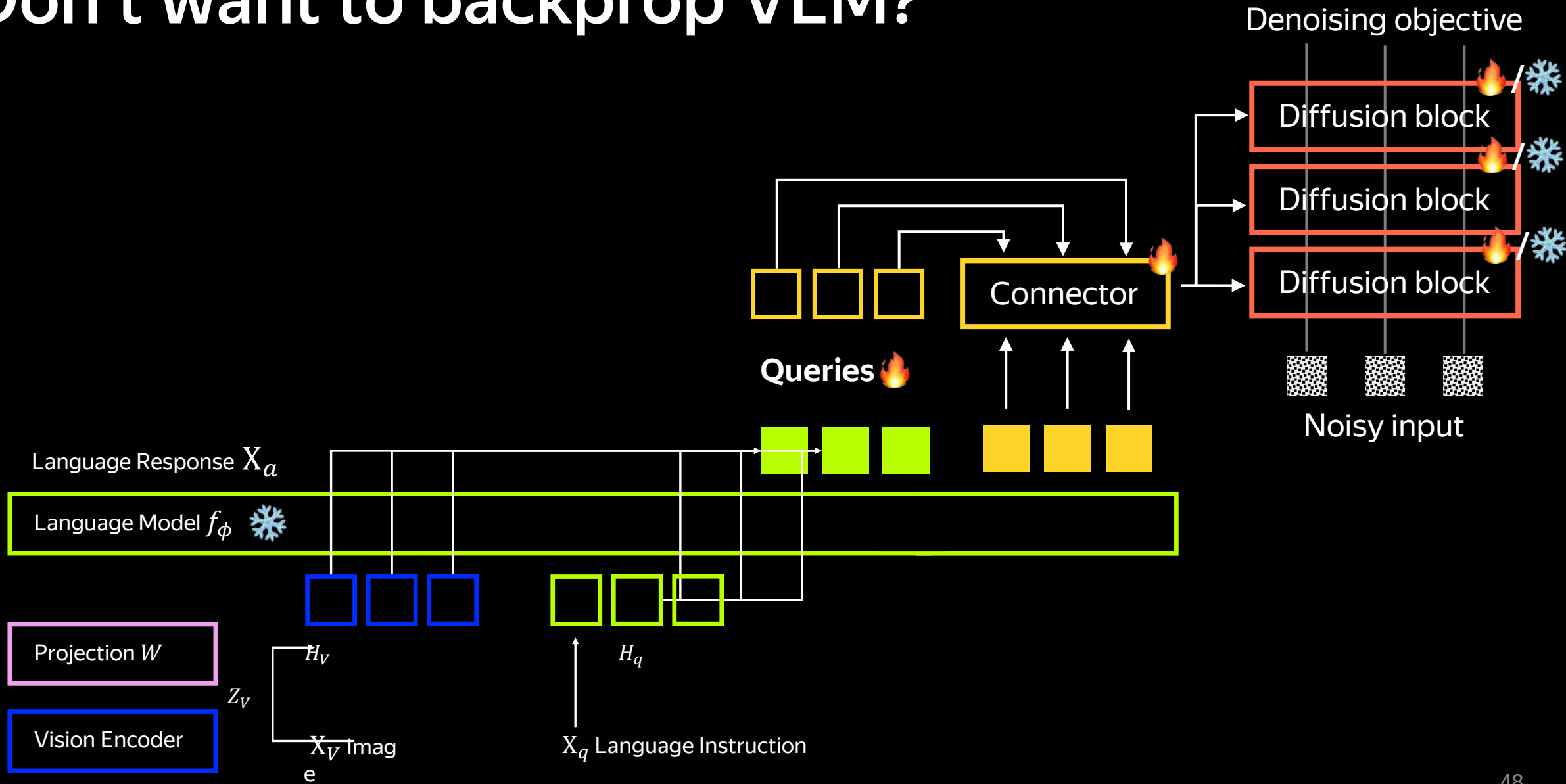
Baseline connection scheme



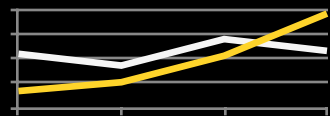
Q-Former and MetaQueries



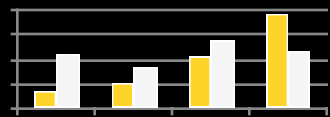
Don't want to backprop VLM?



Recap

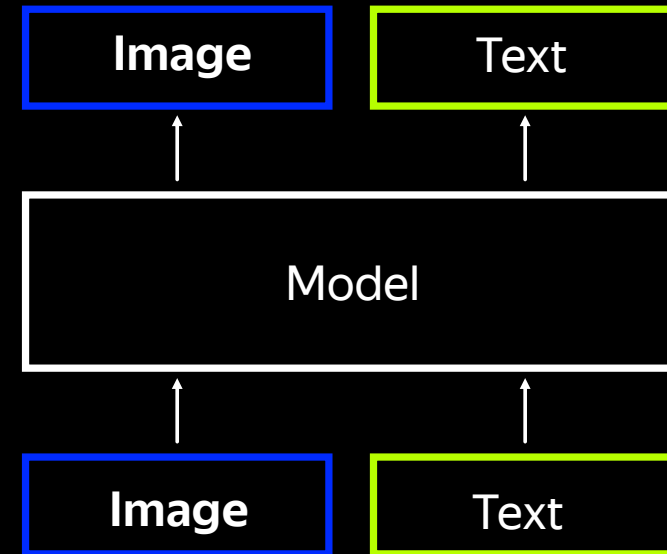


Continuous

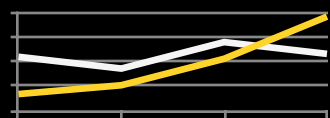


Discrete

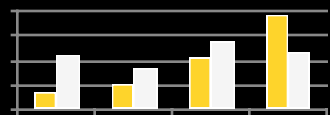
Images representation



Recap



Continuous

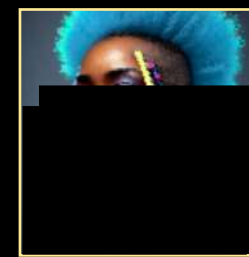


Discrete

Images representation



Diffusion



Autoregression

Loss

Image

Text

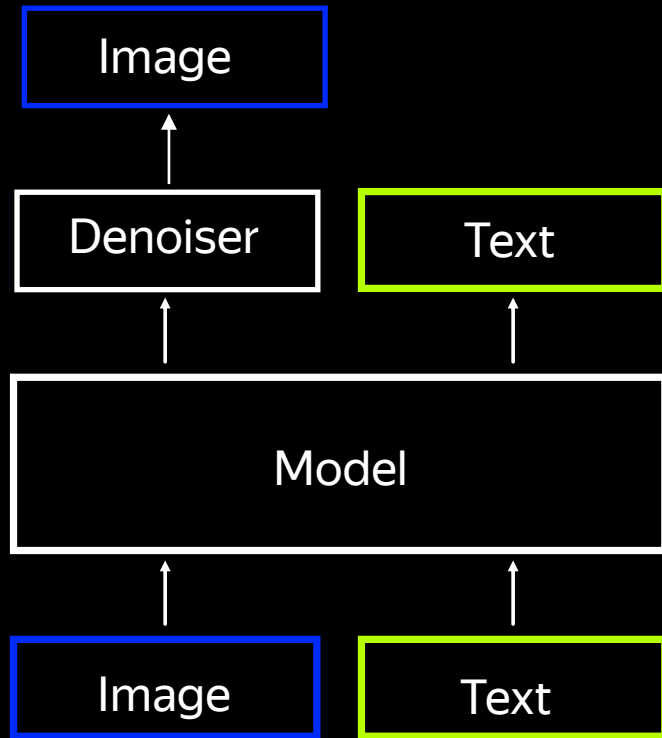
Model

Image

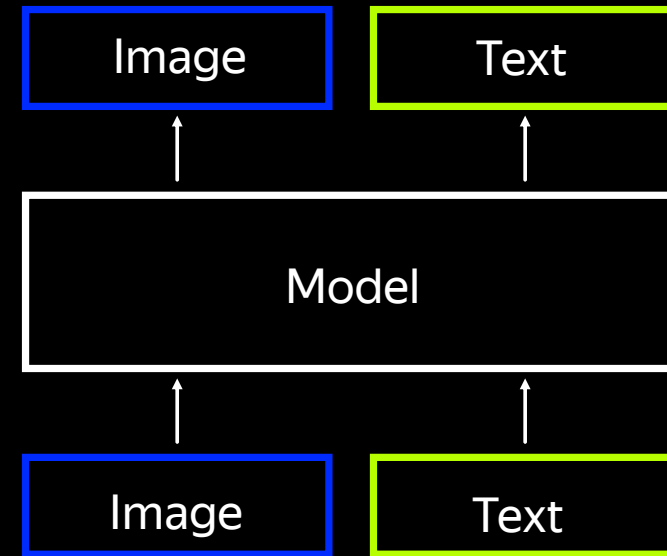
Text

Recap

Image gen module



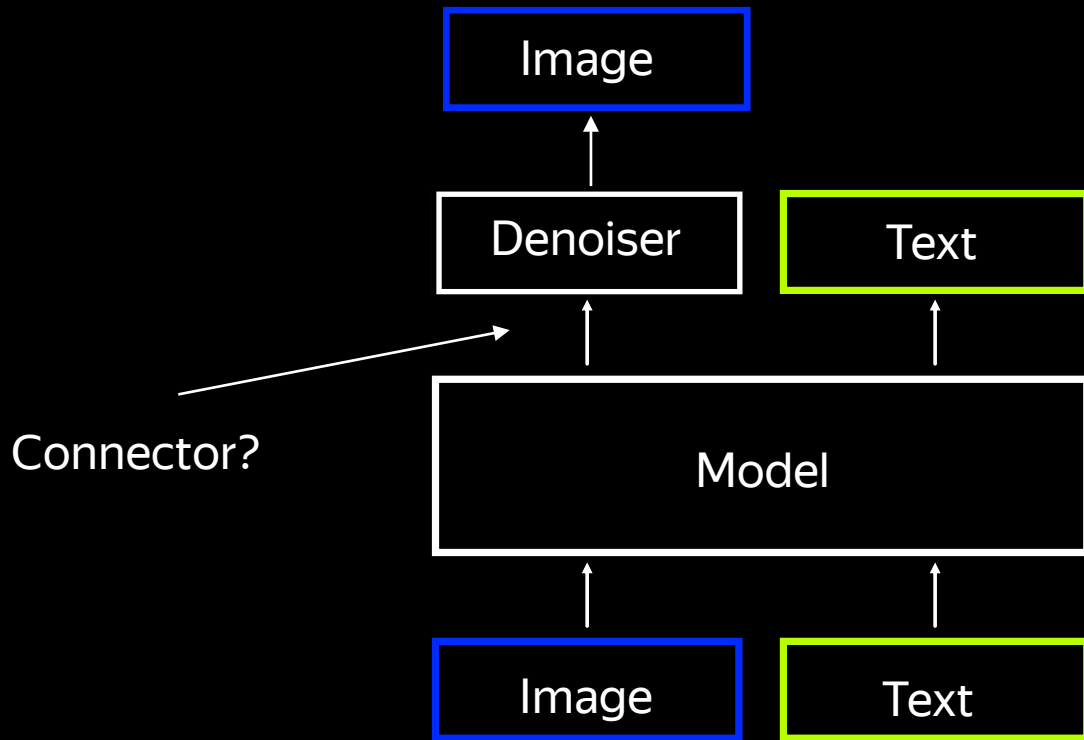
External



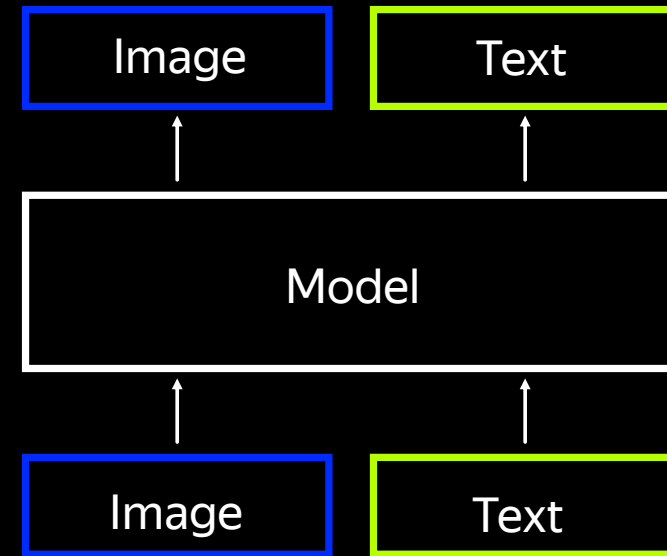
Integrated

Recap

Image gen module

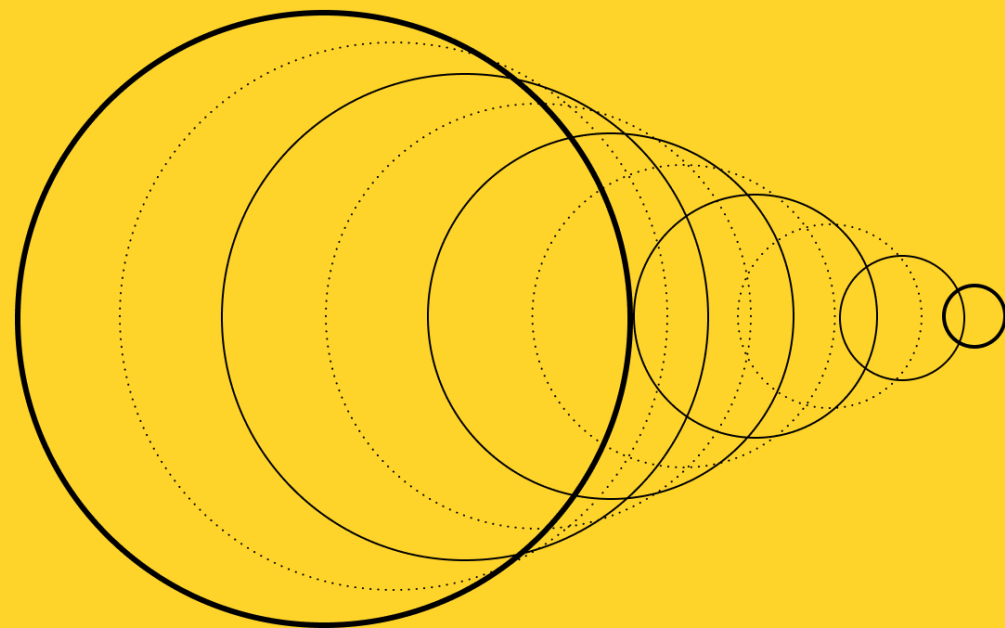


External



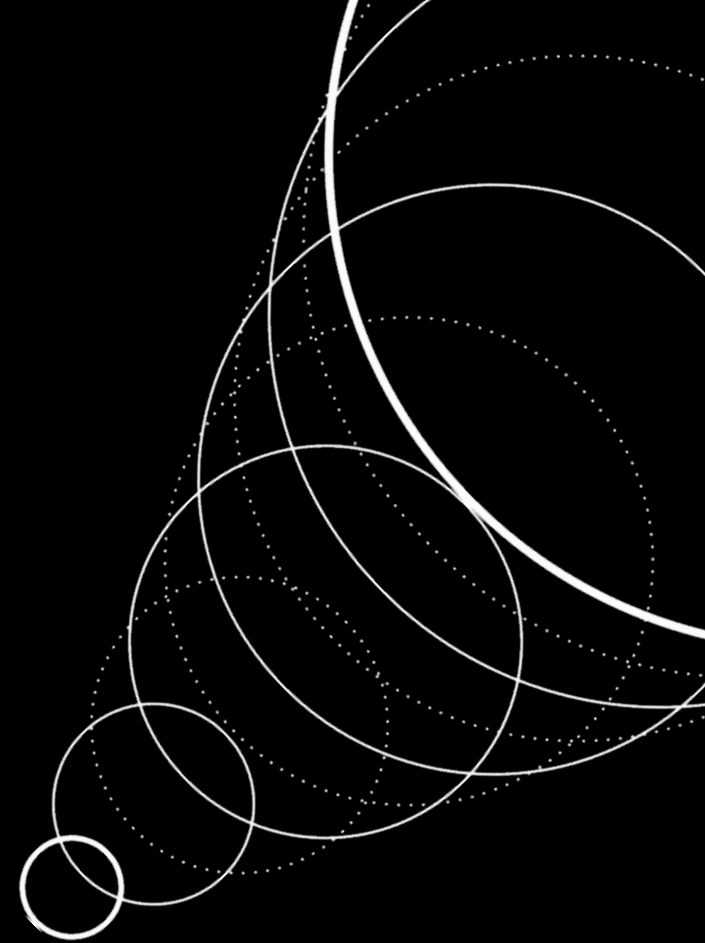
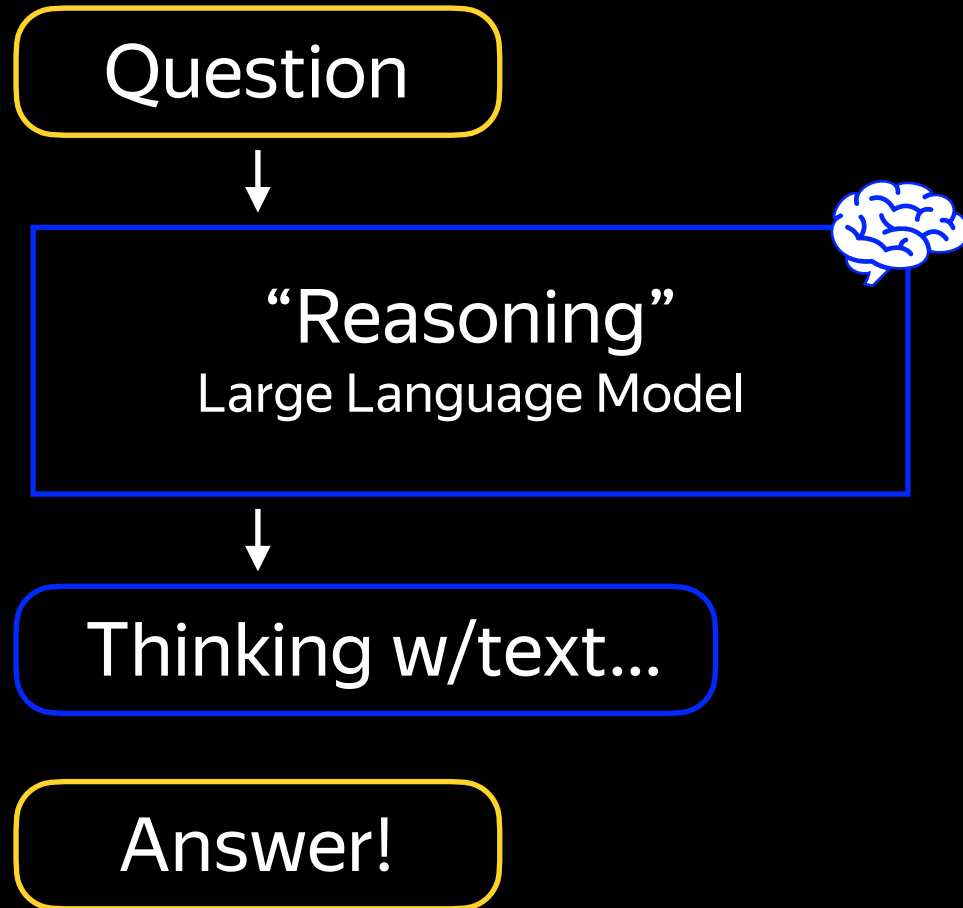
Integrated

Yandex Research

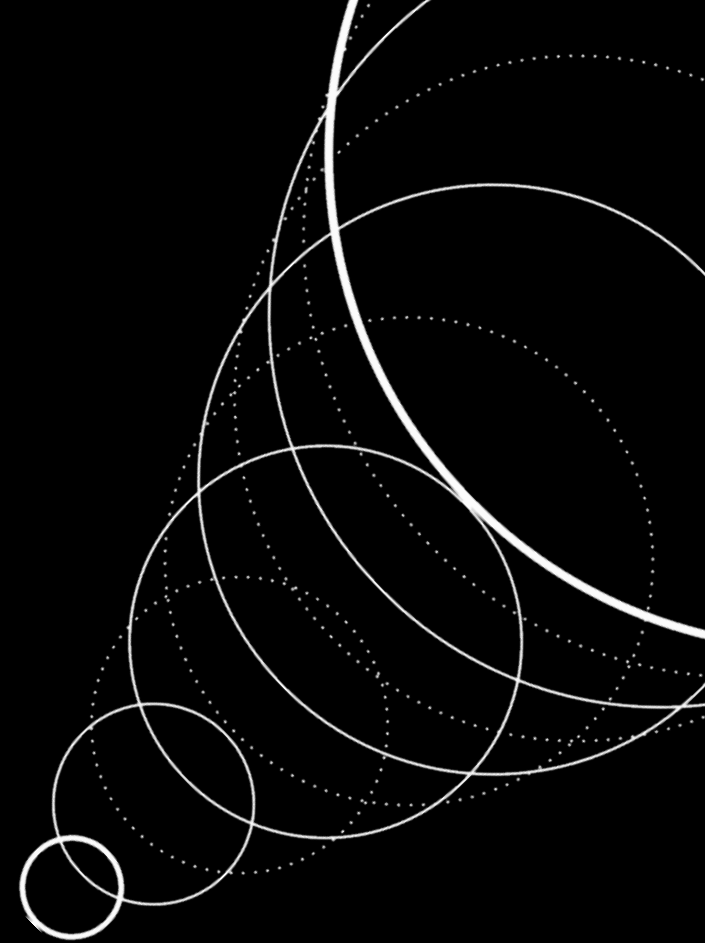
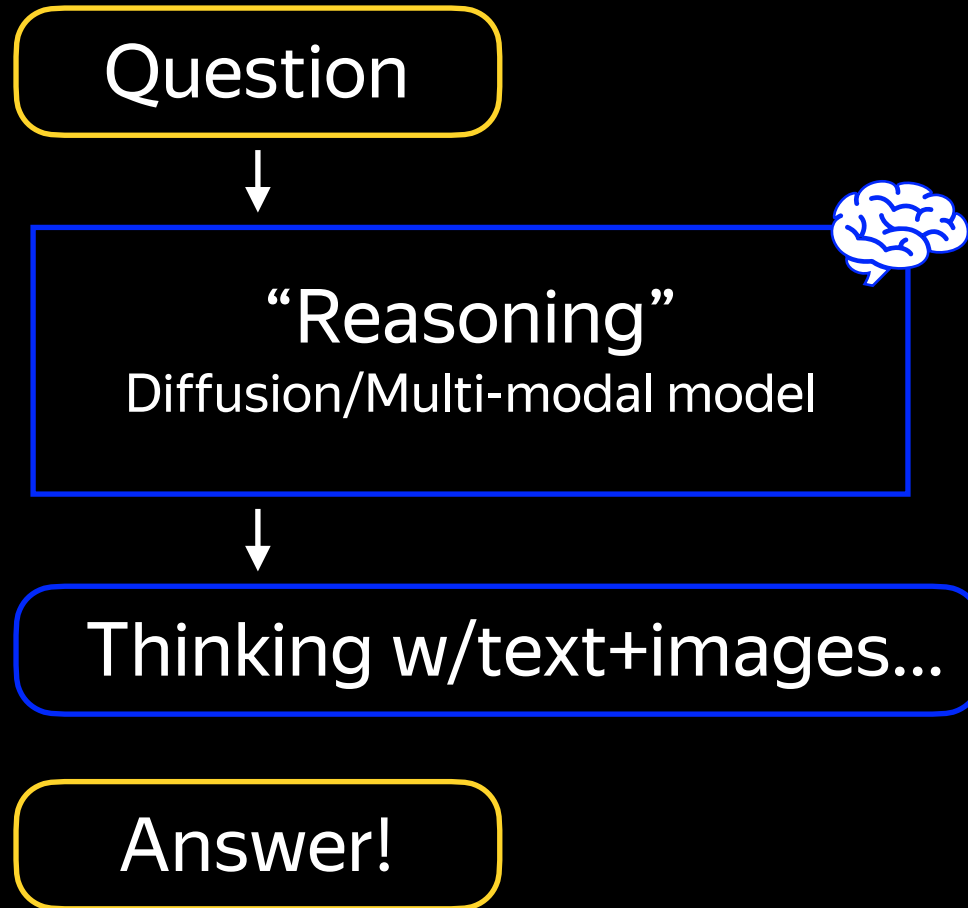


Inference-time Compute Scaling

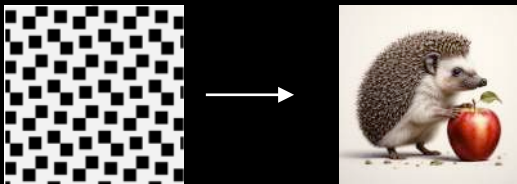
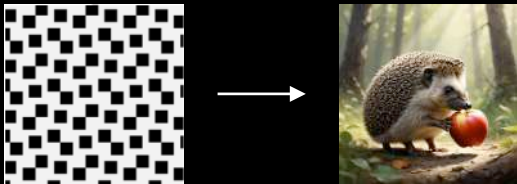
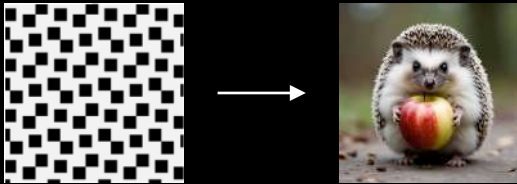
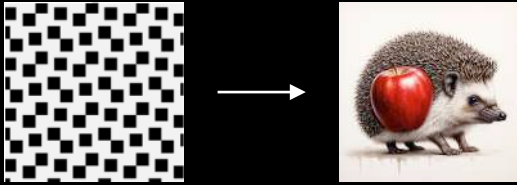
Very much known for LLMs



Very much known for LLMs

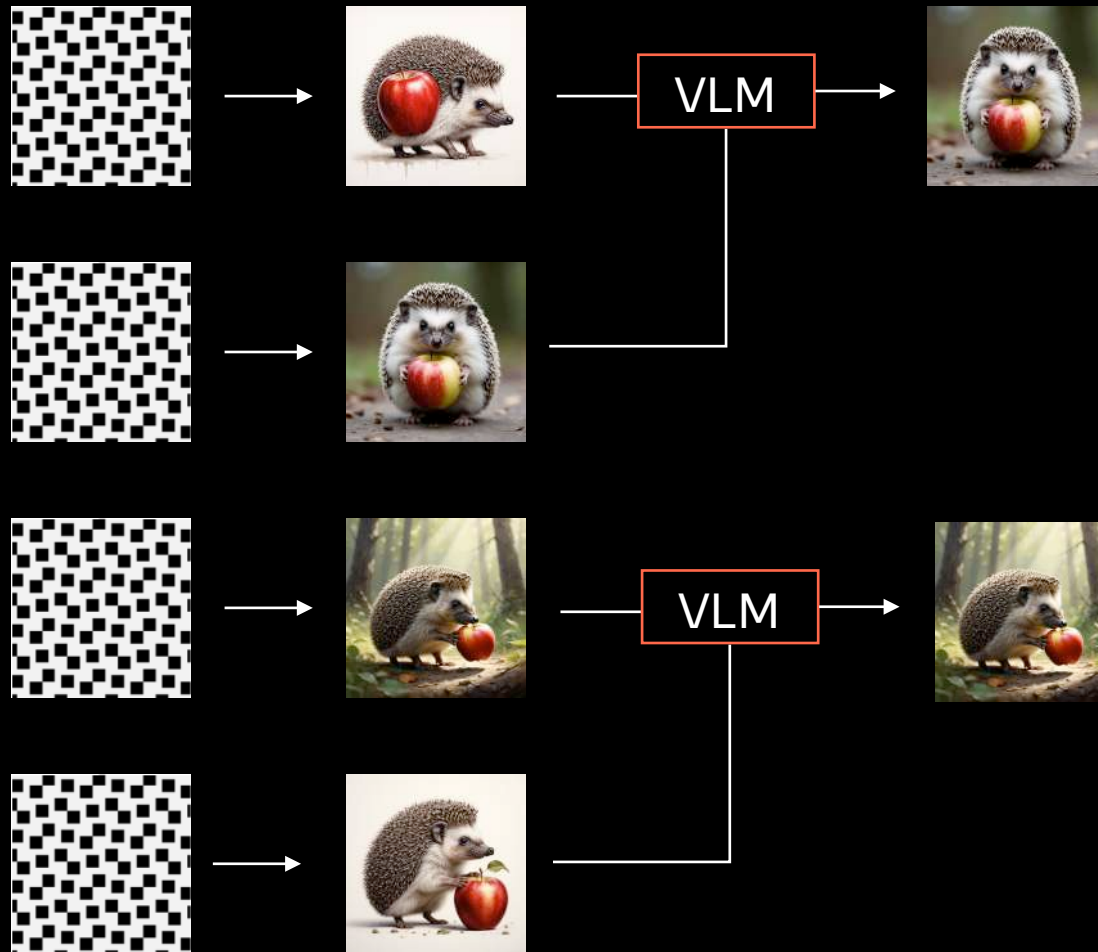


Best-of-N baseline



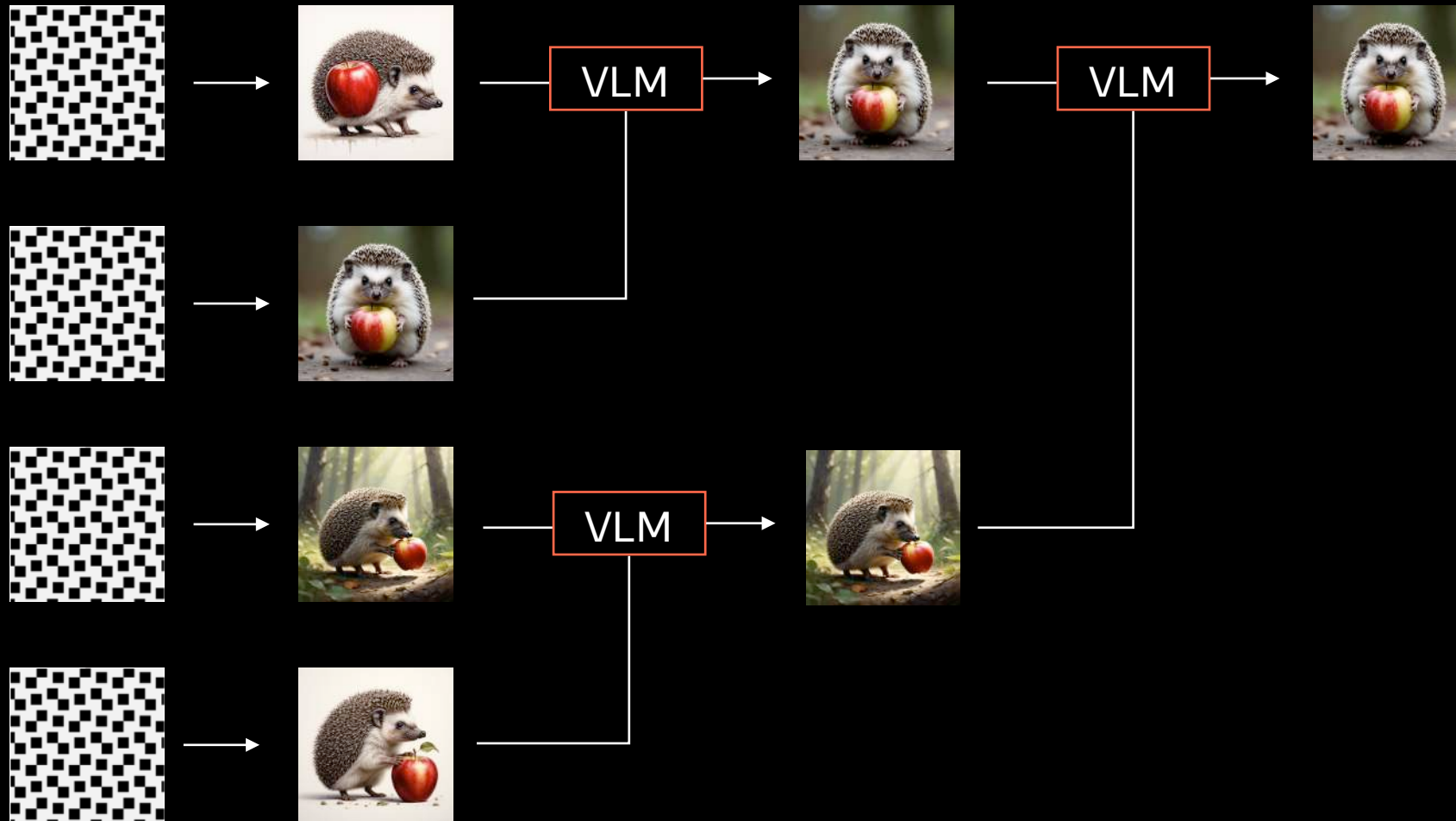
“hedgehog with an apple”

Best-on-N baseline



"hedgehog with an apple"

Best-on-N baseline

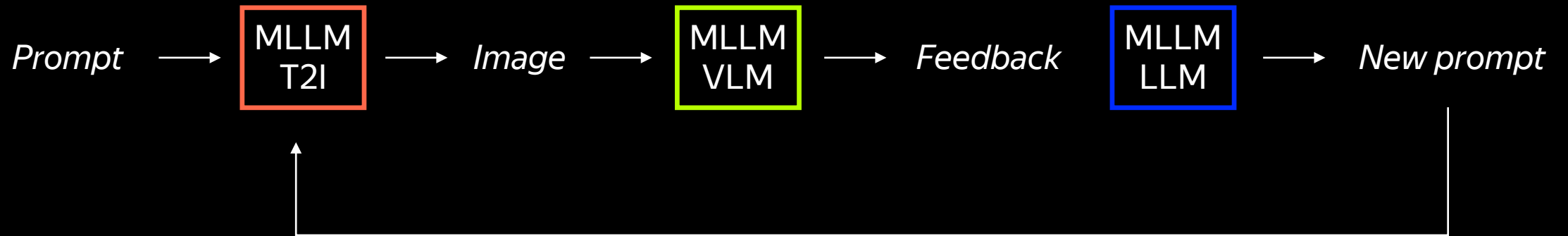


"hedgehog with an apple"

External VLM-as-judge



Can multi-modal model do all that itself?



Boosts text gen and `many objects`

No TTS

TTS

“do an inscription `rash 435`”

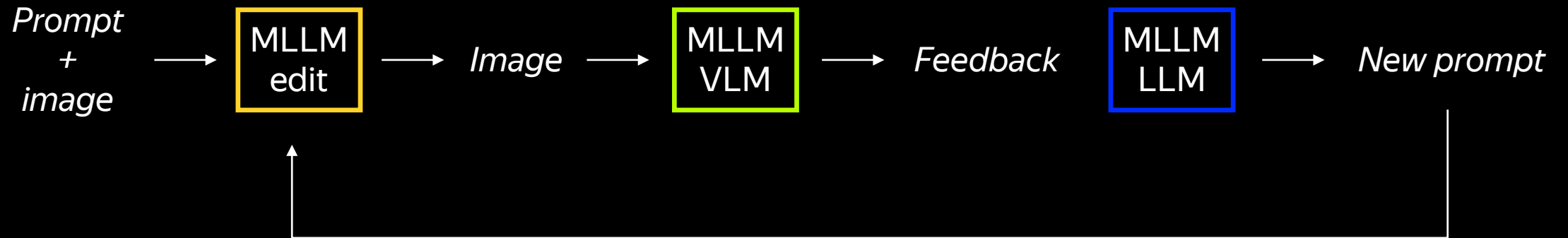


“draw 5 dogs of different colors”



* preliminary results

Can we do the same for editing?



Also works pretty well

“change the action of the horses to galloping”



No TTS



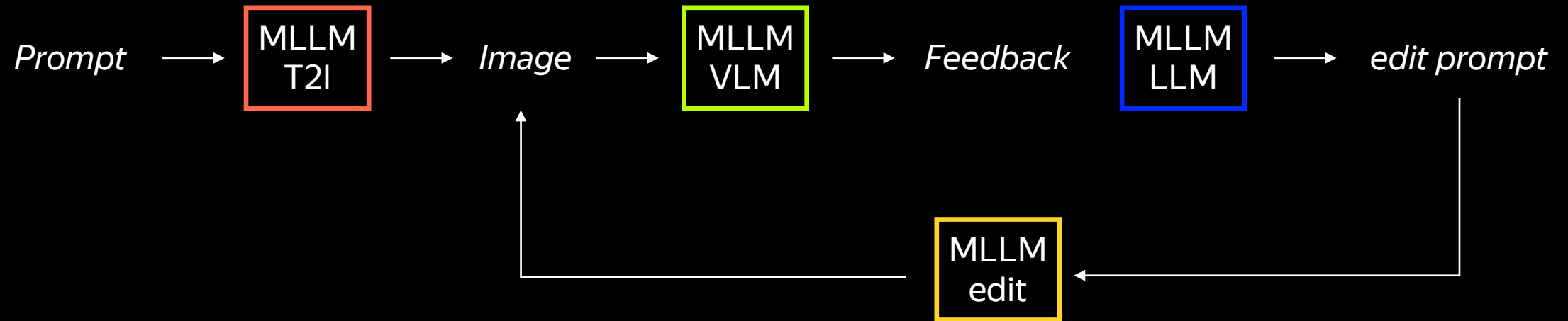
TTS



“Extract the person from the photo and dress them in a police uniform”



Use all four tasks?



1



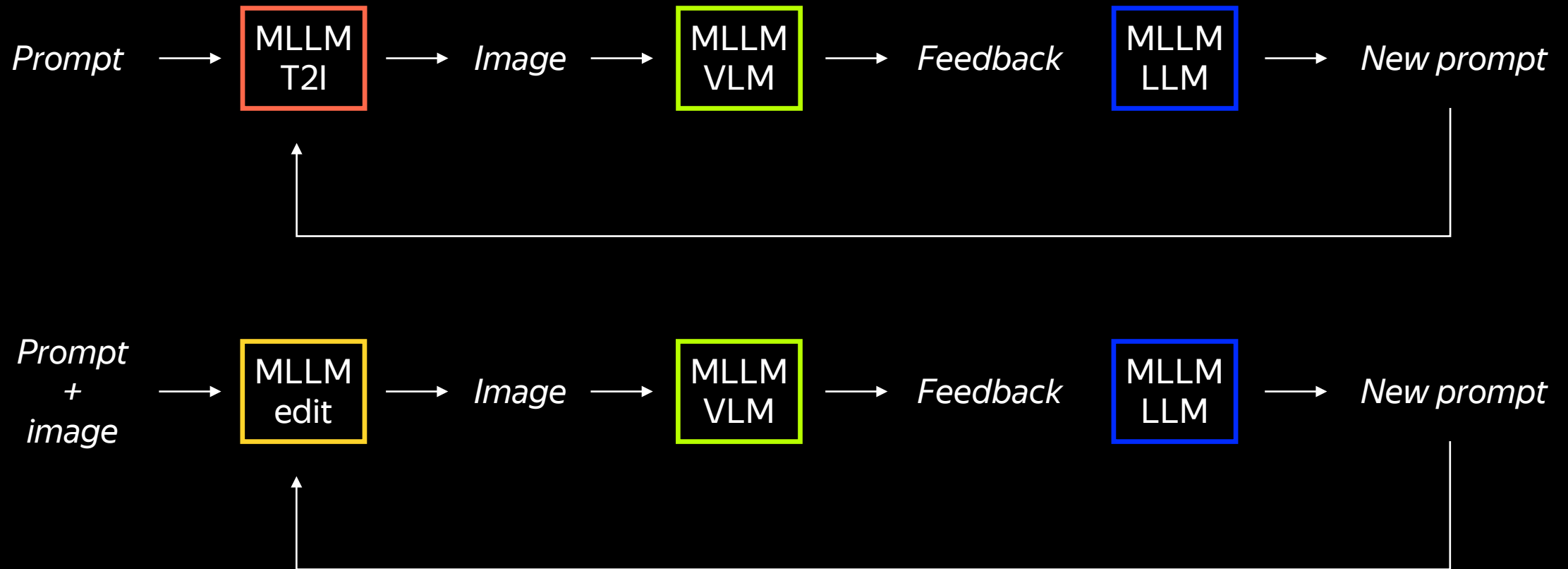
3



5



Distill again?



What stops us from training a reasoner on these synthetic chains? 66

Open Questions

Source of actual quality gain?

- Does the VLM know text-image relations better than T2I?
- If so, why? (gen objective, architecture, data)
- How to enrich T2I or I2I with that?

Explicit reasoning or distill?

- Does actual reasoning emerge?
- Tasks that require visual reasoning?
- T2I to boost VLM reasoning?

